HOLY CROSS COLLEGE

Life to the Full

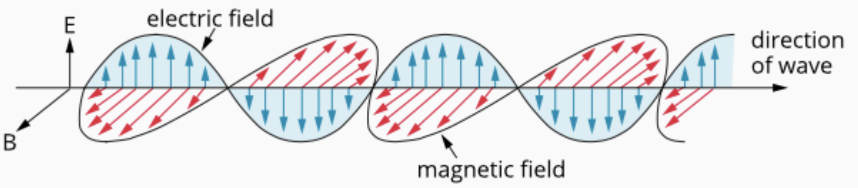
12 PHYSICS

WAVE-PARTICLE DUALITY
&
QUANTUM THEORY

ELECTROMAGNETIC SPECTRUM

The English physicist James Clerk Maxwell proposed that if a changing electric field is produced, for example by a charged particle moving backwards and forwards, then this changing electric field will produce a changing magnetic field at right angles to it. The changing magnetic field would, in turn, produce a changing electric field and the cycle would be repeated.

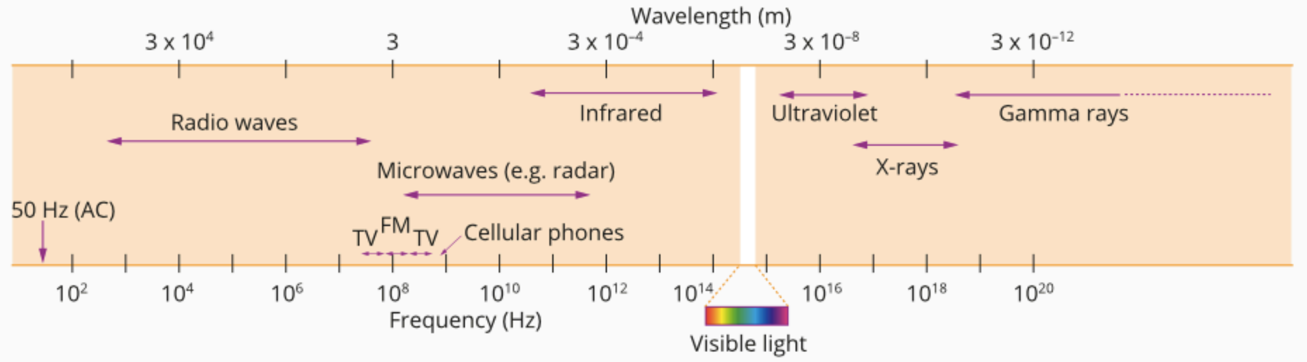
Any vibrating or oscillating electric charge will produce oscillating electric and magnetic fields that are at right angles to each other. It is these fields spreading at the speed of light that we call electromagnetic radiation.



7.3.2 • The electric and magnetic fields in electromagnetic radiation are perpendicular to each other and are both perpendicular to the direction of propagation of the radiation.

The source of the electromagnetic radiation is usually an oscillating electric charge.

The Electromagnetic Spectrum



7.3.3 • The electromagnetic spectrum.

1. ***Power Frequencies***

- Produced by alternating current (AC) generators.
- Emitted by generators and transmission lines.

2. ***Radio Waves***

- Emitted by electrons which are forced to undergo oscillations in a wire.
- The frequency of oscillation governs the frequency of the radio waves.
- If radio waves intercept a conductor of appropriate length (1/2 of the wavelength of the wave) they start electrons in the conductor oscillating at the same frequency. This is the basis of an aerial or antennae.

3. ***Microwaves***

- Generated in magnetrons and klystron tubes where electrons are made to resonate in a small cavity.
- Used in radar, communications and microwave ovens.

4. ***Infra-Red***

- Produced by excited outer electrons in atoms and molecules returning to their ground state.
- Felt as heat on the skin.

5. ***Visible***

- Produced as per infra-red.
- Allows animals to see objects.

6. ***Ultra-Violet***

- Produced as per infra-red.
- Produces tanning/skin cancers in humans, allows plants to photosynthesise, important in the formation of photochemical smog.

7. ***X-Rays***

- Produced by rapidly decelerating electrons after hitting a metal target, as well as inner electrons of atoms returning to their ground state.
- Useful for diagnosis of medical problems inside the body.

8. ***Gamma Rays***

- Produced by the oscillation or acceleration of charges within the nuclei of atoms.
- Useful in industry to detect problems with structures involving steel and concrete.

Dual Nature of Light

The present theory of light is a mixture of two competing theories - the ***electromagnetic theory*** (where light shows wave-like properties) and the ***quantum theory*** (where light shows particle-like properties).

Electromagnetic Theory

Light can be reflected, refracted and diffracted just like water waves, hence it was (before 1900) considered wave-like.

- See Ch. 7.1 of Pearson Physics 12 WA and review the following.
  - Huygen's Principle - role of secondary wavelets.
  - Reflection and refraction.
  - Diffraction and interference.
- Revise any notes from Year 11 on this section.

Quantum Theory

This explained other observed phenomena such as the ***photoelectric effect***.

Electrons can be removed from certain metallic substances by light. To do this, the light is thought to be made up of "packets" or "quanta" of energy called ***photons***.

Thus, it appears light is made up of "particles".

It is accepted by scientists that light behaves as waves or particles, depending on the experiment being performed.

EVIDENCE SUPPORTING ELECTROMAGNETIC THEORY

(See Ch. 7.2 of Pearson Physics 12 WA.)

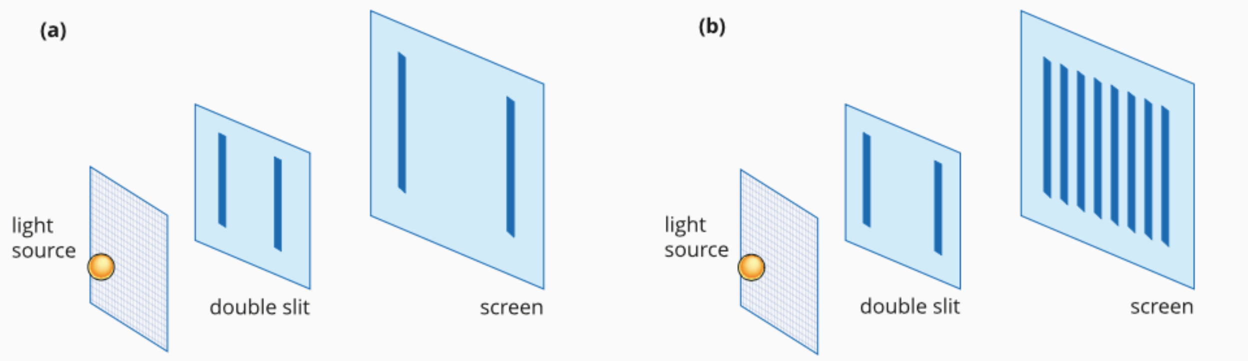
Young's Double-slit Experiment

Between the 17th and 19th centuries, most scientists considered light to be a stream of particles. This idea was based on the 'corpuscular' theory proposed by Sir Isaac Newton.

In 1803, an English scientist called Thomas Young performed an experiment in which he shone ***monochromatic light*** on a screen containing two very tiny pinholes. On the far side of the double-pinhole screen, he placed another screen on which he observed the pattern produced by the light passing through the pinholes.

According to the particle theory, light should have passed directly through the pinholes to produce two bright spots on the screen.

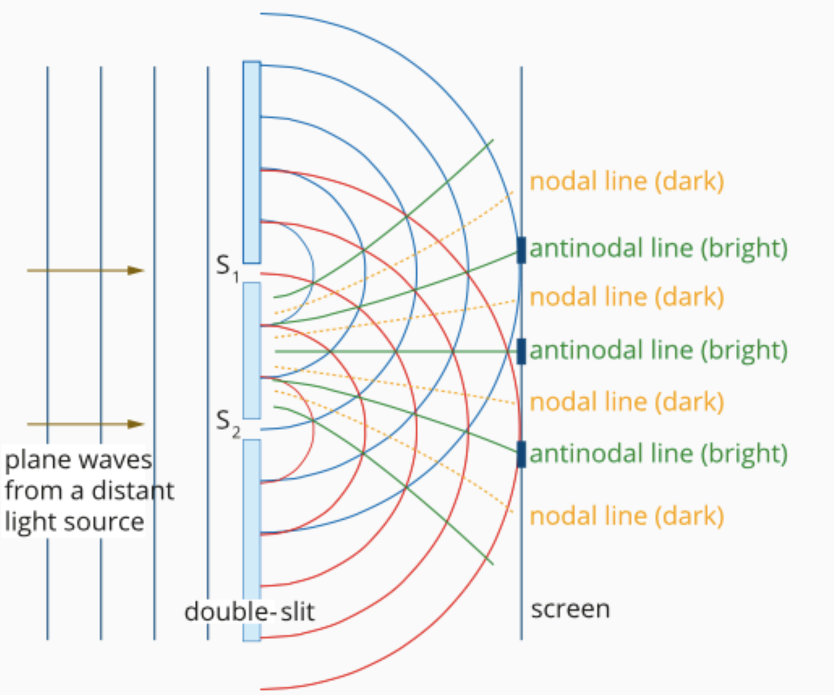
Instead, Young observed a series of bright and dark fringes.



7.2.3 • The particle theory of light predicted that Young's experiment should produce two bright spots (a). But his actual experiment (b) produced a series of bright and dark fringes.

Young was able to explain this pattern by treating light as a wave. He assumed that the monochromatic light was like plane waves and that, as they passed through the narrow slits, these plane waves were diffracted into circular waves as shown below.

The interference pattern produced by these two waves would result in lines of constructive (antinodal) and destructive (nodal) interference that would match the bright and dark fringes respectively.



7.2.4 • The interaction of two circular waves can produce a pattern of antinodal (constructive interference) and nodal (destructive interference) lines.

Resistance to the Wave Model

Young's wave explanation for his experiment was not immediately accepted by the scientific community. Many scientists still preferred the corpuscular (particle) theory of Newton.

In 1818, Augustin-Jean Fresnel was able to provide a mathematical explanation for Young's double-slit experiment.

Simeon Poisson, a passionate believer in the particle model, argued that if Fresnel's mathematics was applied to a light shining on a round disc, then there should be a bright spot observed in the middle of a shadow. As no one has observed this, then Poisson believed the wave model was incorrect.

A colleague of Poisson performed the experiment and did see the bright spot. Hence the wave model was almost universally accepted, and the diffraction pattern was named the "Poisson bright spot".

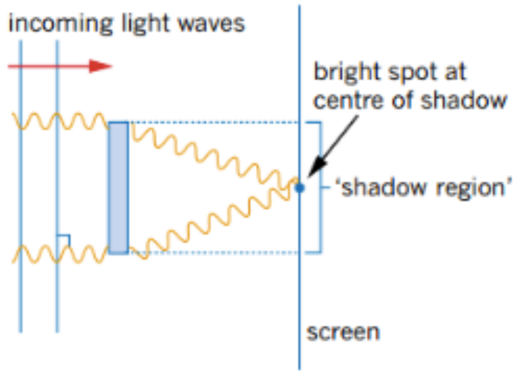


FIGURE 7.2.9 Waves of light incident on a solid disc diffract to give a point of light in the centre of the shadow zone. This is convincing evidence for the wave nature of light.

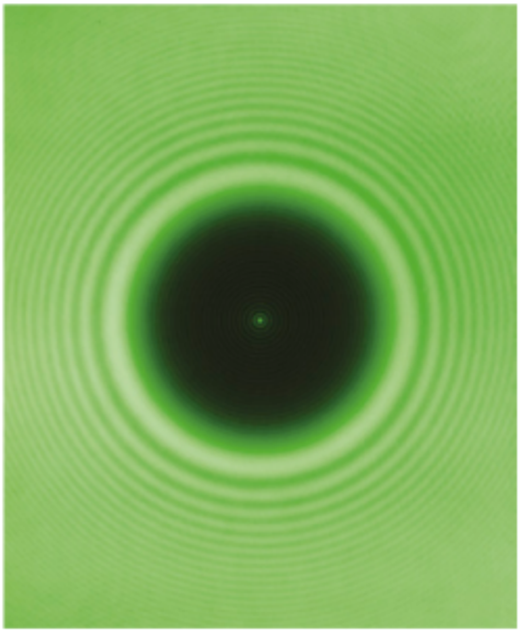


FIGURE 7.2.10 The bright spot inside the shadow region of this image is caused by the diffraction and interference of light waves. The image also shows diffraction and interference patterns surrounding the shadow.

EVIDENCE SUPPORTING QUANTUM THEORY

(See Ch. 7.4 of Pearson Physics 12 WA.)

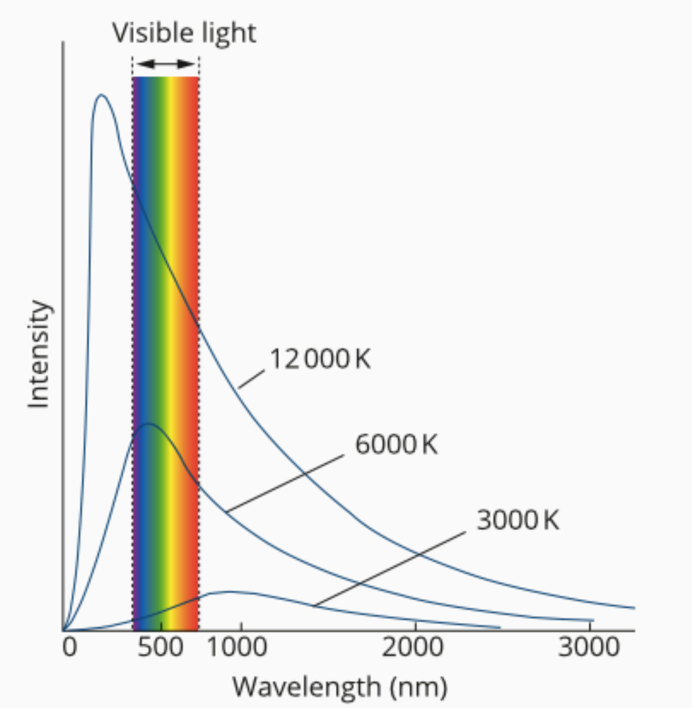
At the turn of the of the 20th century, a number of scientists turned their attention to phenomena such as black-body radiation and the photoelectric effect that could not be readily explained using Maxwell's electromagnetic wave model. The study of these phenomena required the development of much more sophisticated models for light and eventually led to a revolution in our understanding of the nature of energy and matter.

Black-body Radiation

All objects that have temperatures above absolute zero emit electromagnetic radiation. For objects at room temperature, practically all of this radiation is emitted in the infrared part of the electromagnetic spectrum, which is not visible to the human eye.

At higher temperatures, not only is the total amount of radiation emitted larger, but also a higher proportion of it is in the visible part of the spectrum.

- ***Around 3000 K*** - a significant amount of energy is emitted as ***red wavelengths*** of the visible spectrum and the object appears 'red hot'.
- ***Around 6000 K*** - energy is radiated roughly evenly across different wavelengths of the ***visible spectrum*** so the object appears 'white hot'.
- ***Around 12000 K*** - more of the energy is being radiated in the ***ultraviolet and violet parts of the spectrum***, so the object appears 'blue hot', like the hottest flame of a Bunsen burner.



7.4.6 • Objects at 3000 K, 6000 K and 12 000 K have different emission spectra. As the temperature increases, the amount of energy radiated (shown by the area under these graphs) increases, and the peak wavelength decreases (it becomes more blue).

A ***black-body*** is an ideal absorber and absorbs all incident radiation; it is in equilibrium with its surroundings, as ***it radiates and absorbs energy at the same rate***, and so its ***temperature remains constant***.

A blackbody can be approximated by the inside of a cavity with a small entrance hole, because all light entering the hole is trapped.

The shape of the radiation spectrum only depends on the temperature of the black-body, not its colour.

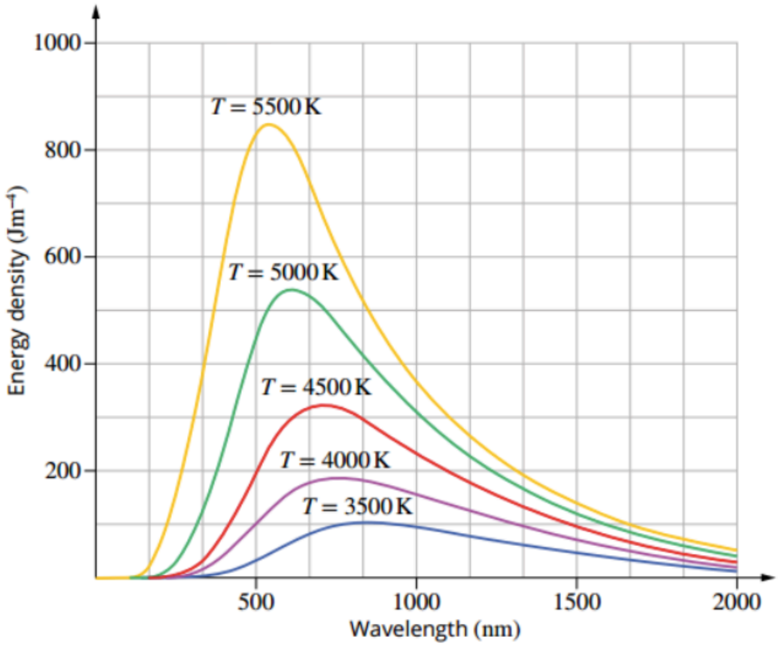


FIGURE 7.4.5 Blackbody emission spectrum for temperatures of 3500 K up to 5500 K.

As the temperature increases, the peak in energy density increases and the total energy emitted (***represented by the area under the curve***) by the object increases.

In addition, the peak wavelength (λ_{max}) decreases; this corresponds to an increase in peak frequency and energy.

This shift to shorter wavelengths was found to have the following relationship, known as ***Wien's Displacement Law***.

$$\boxed{\lambda_{max}T = 0.2898 \times 10^{-2} \text{ mK}}$$

where T = the temperature in kelvin (K)
λ_{max} = the wavelength (m)

Classically, thermal radiation is emitted by accelerating charges near the surface of a material. The charges have a distribution of accelerations that leads to a range of thermal energies.

The classical expression, known as the ***Rayleigh–Jeans Law***, gives the average energy per oscillating or vibrating charge proportional to temperature.

The radiated energy can be considered to be produced by ***standing waves or resonant modes*** within the cavity.
At long wavelengths (λ), there is a reasonable agreement between the experimental data and the calculation.

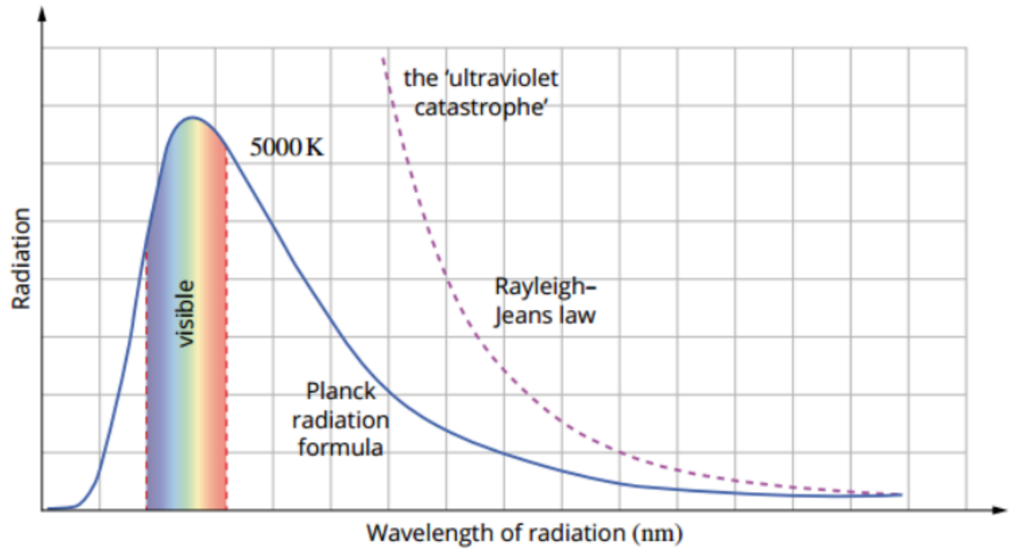


FIGURE 7.4.6 Comparison of the experimentally observed blackbody radiation intensity and the Rayleigh–Jeans classical calculation.

However, as the wavelength gets shorter (frequency increases), the number of modes gets significantly larger and the energy density increases and approaches infinity. This is known as the ***ultraviolet catastrophe***.

Although scientists had developed equations to explain the total amount of energy emitted by a black-body and the wavelength peak of the emission spectrum, these could not be adequately explained using the wave model of light.

Planck's equation

In 1900, the German physicist Max Planck was studying the spectrum for light emitted by a black-body when he realised that the shape of the spectrum resembled a statistical distribution curve, like the normal distribution curve used to study random samples of populations.

In order for the theory to fit experimental observations, Planck incorporated two fundamental postulates.

1. Molecules vibrate at discrete energies or frequencies, as show below.

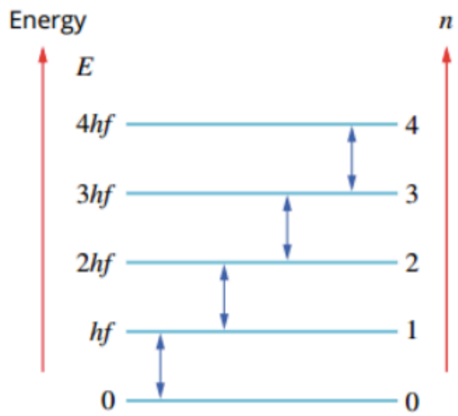


FIGURE 7.4.7 Molecules vibrate at discrete energies, $E_n = nhf$. Molecules gain energy by absorbing a photon (upwards arrow) and lose energy by emitting a photon (downwards arrow). The photons emitted are typically low energy, in the infrared at room temperature.

2. Molecules absorb and emit radiation in in discrete packets called ***photons***, and the energy of an individual photon is $E = hf$.

Using this insight, Planck was able to show that if he assumed light was emitted as ***discrete packets, each with an energy proportional to its frequency***, then he could exactly predict the black-body emission spectrum.

Planck called the discrete packets of energy ***quanta*** and developed an equation for the energy of each quantum:

$$\boxed{E = hf}$$

where f = the frequency of the electromagnetic radiation (Hz)
h = Planck's constant ($= 6.63 \times 10^{-34}$ Js)

Since electromagnetic radiation is more commonly described according to its wavelength, scientists often combine Planck's equation with the wave equation $c = f\lambda$ as follows.

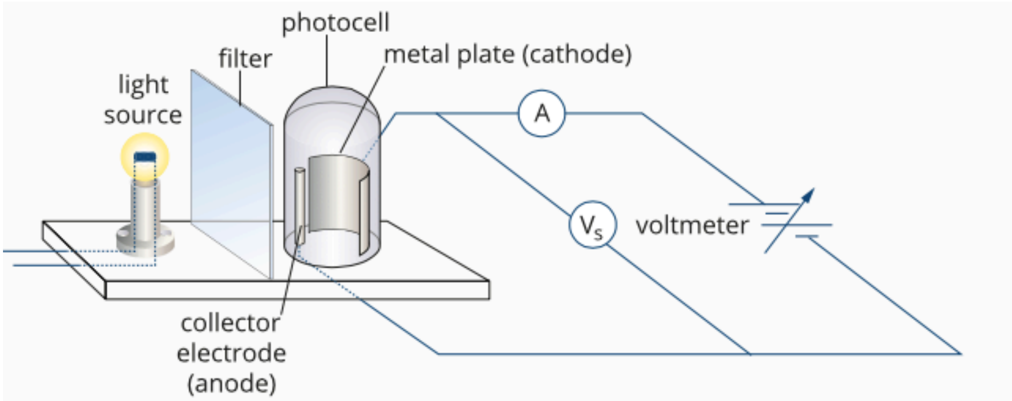
$$\boxed{E = hf = \frac{hc}{\lambda}}$$

Photoelectric Effect

Another phenomenon that could not be explained using the wave model for light was known as the photoelectric effect.

Scientists noticed that when some types of electromagnetic radiation shine on a piece of metal, the metal becomes positively charged. This positive charge is due to electrons being ejected from the surface of the metal. The electrons became known as ***photoelectrons*** because they were released due to light or other forms of electromagnetic radiation.

A common apparatus used to observe the photoelectric effect is shown in the figure below. A clean metal surface (the cathode) is illuminated with light from an external source. If the light causes photoelectrons to be emitted, they are detected at the anode. The flow of electrons is called the ***photoelectric current*** and is registered by a sensitive ammeter.



7.4.8 • In 1916, Robert Millikan used an apparatus similar to this to investigate the photoelectric effect. He found that for different metals, a different minimum frequency was required in order to produce photoelectrons.

The circuit includes a variable voltage supply, which can be used to make the cathode negative (and the anode positive). When this is done, the photoelectrons will be helped by the resulting electric field to the anode, and a maximum possible current will be measured.

Alternatively, the voltage may be adjusted to make the cathode positive and the anode negative.

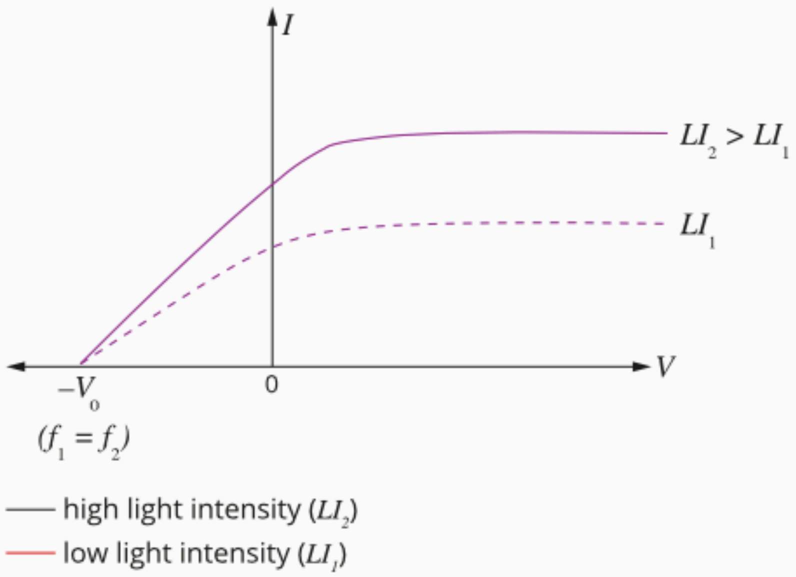
The German physicist Philipp Lenard made a number of surprising discoveries about the photoelectric effect. Lenard used the filter to vary the frequency of the incident light. He discovered:

- For a particular cathode metal, there is a certain frequency below which no photoelectrons are observed. This frequency is called the ***threshold frequency*** f_o.

- For frequencies of light greater than the threshold frequency (i.e. $f > f_o$), photoelectrons will be collected at the anode and registered as a photoelectric current.

- For frequencies below the threshold frequency (i.e. $f < fo$), no photoelectrons will be detected.

Lenard also discovered that, for light whose frequency is greater than the threshold frequency ($f > f_o$) , the ***rate at which the photoelectrons are produced varies in proportion with the intensity of the incident light*** as shown in the graph below.



7.4.9 • Photoelectric current (I) plotted as a function of the voltage (V) applied between the cathode and the anode in the photocell for different light intensities. For brighter light of the same frequency, there is a higher photoelectric current, but the same stopping voltage.

This graph shows a number of important features of the photoelectric effect.

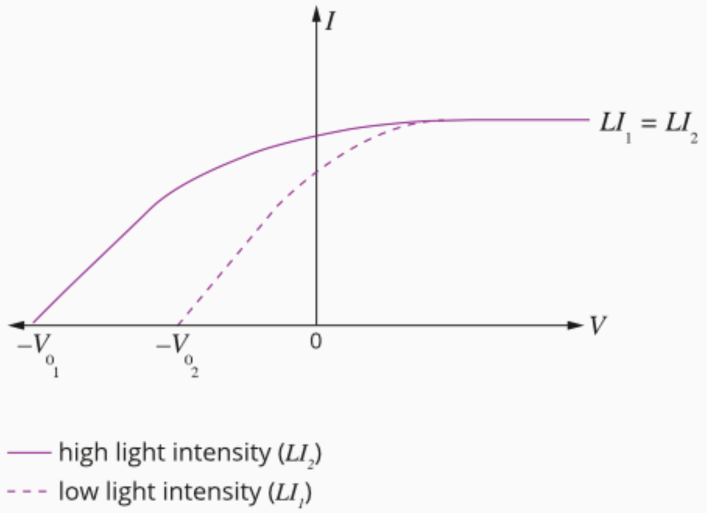
- When the light intensity increases, the amount of photoelectric current increases.

- When the applied voltage is ***positive***, photoelectrons are attracted to the collector electrode. A small positive voltage is enough to ensure that every available photoelectron is collected and the current reaches its maximum value.

- When the applied voltage is ***negative***, photoelectrons are repelled by the collector electrode and the photoelectric current is reduced.

  There is a voltage, $-V_o$, for which no photoelectrons reach the collector. This is known as the ***stopping voltage***. For a particular frequency of light on a particular metal, this stopping voltage is a constant.

When light sources of the same intensity but ***different frequencies*** are used, they produce the same maximum current, but the ***higher frequency light has a higher stopping voltage*** (see figure below).



7.4.10 • Photoelectric current (I) plotted as a function of the voltage (V) applied between the cathode and the anode in the photocell for different frequencies of incident light with the same intensity. Both frequencies produce the same maximum photoelectric current, however, light with the higher frequency requires a larger stopping voltage.

The stopping voltage V_0 is used to calculate the maximum E_k of the photoelectrons. To overcome the potential difference equal to the braking voltage, an electron needs to perform work W equal to the product of the voltage V and the charge of an electron q.

$$\text{i.e.} \quad \boxed{W = E_{k\,(max)} = V_o\, q}$$

Finally, as long as the incident light has a frequency above the threshold frequency of the cathode material, photoelectrons are found to be emitted without any appreciable time delay. This fact holds true regardless of the intensity of the light.

Explaining the Photoelectric Effect

The characteristics of the photoelectric effect could not be easily explained using a wave model of light.

According to the wave model, the frequency of light should be irrelevant to whether or not photoelectrons are ejected. Since a wave is a form of continuous energy transfer, we would expect that even low frequency light should transfer enough energy to emit photoelectrons if left to shine on the metal for long enough.

Similarly, the wave model predicted that there should be a time delay between the light striking the metal and photoelectrons being emitted as the energy from the wave built up in the metal. There is no delay in the photoelectrons being emitted.

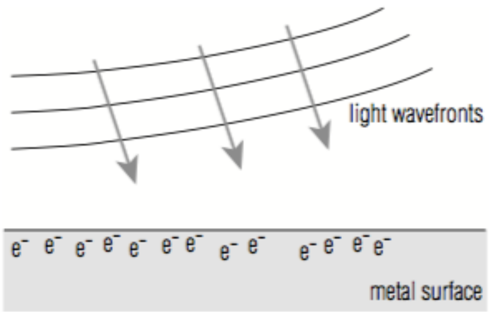


Figure 10.18 The wave model predicts that a wavefront will interact with all electrons near the surface of a material at once. The total energy available will be divided among the electrons. If the light beam has a low intensity, the wave model predicts that there will therefore be a time delay, while energy builds up, before any electrons are ejected from the metal.

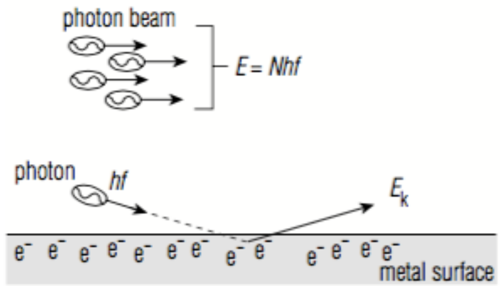


Figure 10.19 The photon model for light suggests that the beam of light carries its energy as a stream of photons, and that each photon can interact with one electron in the metal. A low-intensity beam simply has fewer photons, but each photon still carries the same energy. If individual photons carry sufficient energy then there will be no time delay in emission of an electron when it absorbs a photon.

In 1905, Albert Einstein proposed a solution to this problem. Einstein drew on Planck's earlier work on black-body radiation by assuming that light exists as particles, or 'photons' (like Planck's 'quanta'), each with an energy of $E=hf$. This assumption made the properties of the photoelectric effect relatively easy to explain.

Einstein identified that, for a particular metal, the amount of energy required to produce a photoelectron is a constant value that depends on the strength of the bonding within the metal. This energy was called the ***work function*** *(W)* of the metal.

e.g. The work function of lead is 4.25 eV, which means that 4.25 eV of energy is needed to release one electron from the surface of a piece of lead.

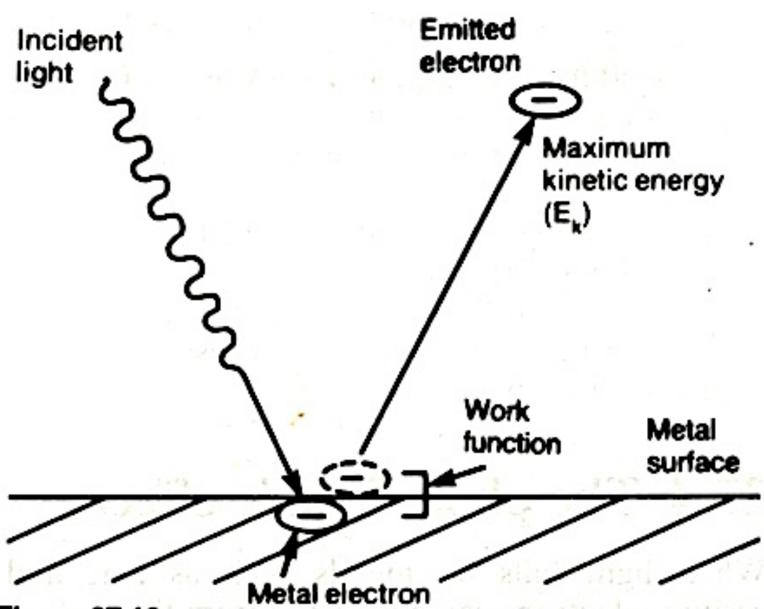


Figure 27.10

Example

If the energy of the photon is less than the work function W, then photoelectrons will not be released.

The photons of violet light (f = 7.5 x 10^{14} Hz) each contain 3.1 eV of energy.

$$\begin{aligned} E &= hf \\ &= (6.63 \times 10^{-34})(7.5 \times 10^{14}) \\ &= 5.0 \times 10^{-19} \text{ J} \\ &= 3.1 \text{ eV} \end{aligned}$$

This means that violet light shining on lead would not produce photoelectrons since the energy of each photon is less than the work function of lead.

This explains why each metal has a threshold frequency - this is the frequency at which the photons have an energy equal to the work function of the metal.

$$\boxed{W = hf_o}$$

If the energy of the photon is greater than the work function of the metal, then a photoelectron is released. Any excess energy in the photon is transformed into the kinetic energy of the photoelectron.

Einstein described this relationship with his photoelectric equation:

$$\boxed{E_k = hf - W}$$

where E_k = kinetic energy of the photoelectron (J)
h = Planck's constant (= 6.63 x 10^{-34} Js)
f = the frequency of the incident light (Hz)
W = work function (J)

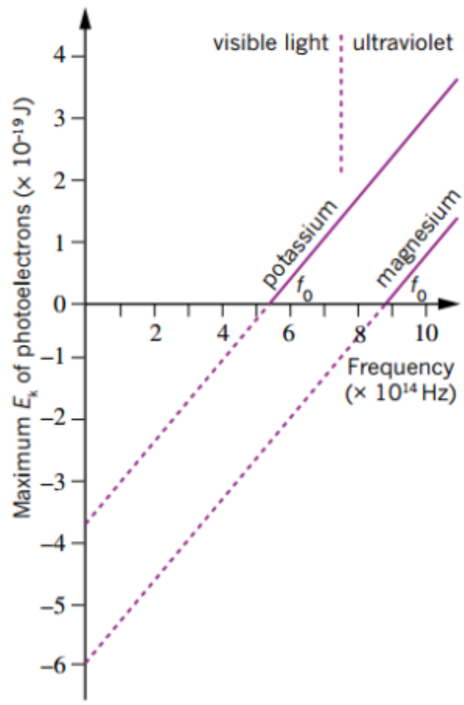


FIGURE 7.4.12 Magnesium has a high threshold frequency that is in the ultraviolet region. The x-intercept gives the threshold frequency, f_0. For potassium, the threshold frequency is in the visible region. The gradient of the graph for each metal is Planck's constant, h. The magnitude of the y-intercept gives the work function, ϕ.

The graphs below show the photoelectric behaviour of different metals. Note the following.

- The gradient is constant for each metal (with a value of Planck's constant h).
- They have different threshold frequencies.

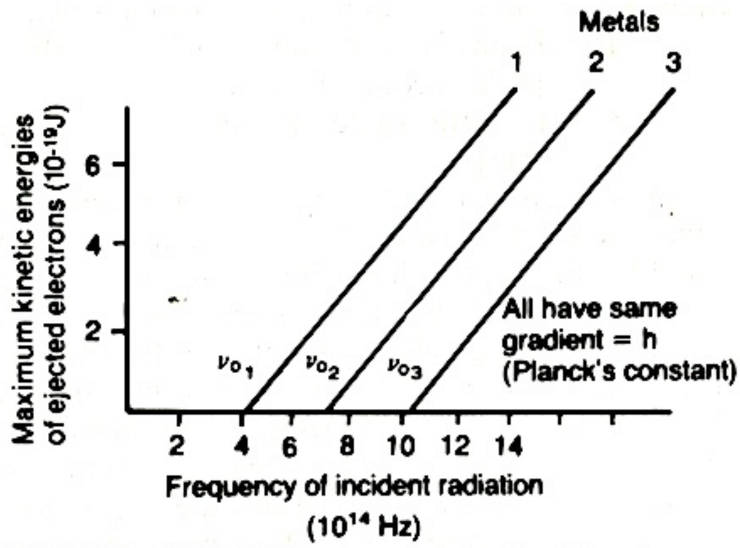
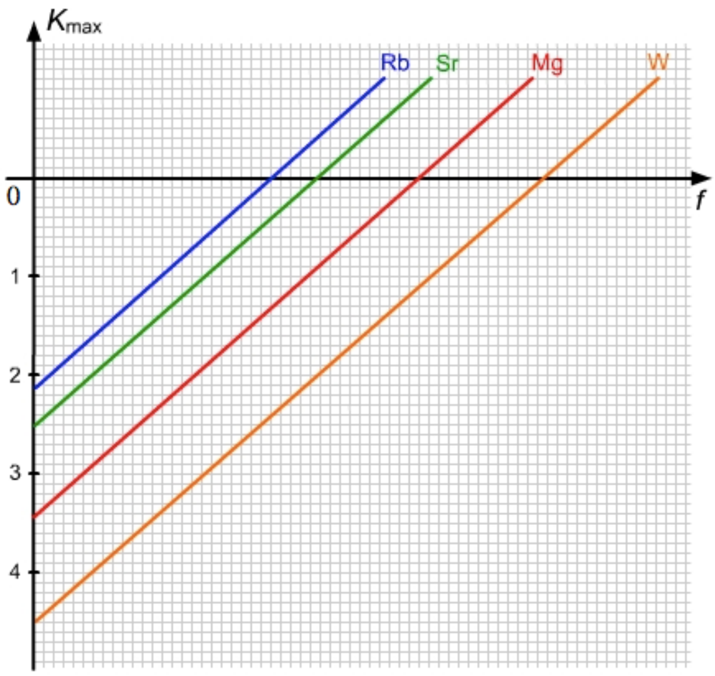


Figure 27.8 Photoelectric characteristics for different metals

By extrapolating the graphs backwards, it is possible to gain some information about the value of the work function W. In the following diagram, E_k values are in eV.

e.g. W (for Mg) = 3.5 eV



Example 1

Note - ϕ is used instead of W for work function.

Example

Electromagnetic radiation of wavelength 1.5×10^{-7} m illuminates a metal surface which has a work function of 6.1×10^{-19} J. What is the energy of the emitted photoelectrons?

Determine the frequency of the incident radiation.

$c = 3.00 \times 10^{8}\ \text{m s}^{-1}$
$\lambda = 1.5\ 10^{-7}\ \text{m}$
$v = ?$

$$\begin{aligned} c &= \lambda v \\ 3.00 \times 10^{8} &= 1.5 \times 10^{-7} \times v \\ v &= 2.0 \times 10^{15}\ \text{Hz} \end{aligned}$$

Determine the energy of the emitted photo electrons.

$v = 2.0 \times 10^{15}$ Hz
$h = 6.625 \times 10^{-34}$ J s
$\phi = 6.1 \times 10^{-19}$ J
$E_k = ?$

$$\begin{aligned} hv &= \phi + E_{k\ max} \\ 6.625 \times 10^{-34} \times 2.00 \times 10^{15} &= 6.1 \times 10^{-19} + E_k \\ 13.2 \times 10^{-19} &= 6.1 \times 10^{-19} + E_k \\ E_{k\ max} &= (13.2 - 6.1) \times 10^{-19} \\ &= 7.1 \times 10^{-19}\ \text{J} \end{aligned}$$

Maximum energy of emitted photoelectrons is 7.1×10^{-19} J.

Example 2

CALCULATING THE KINETIC ENERGY OF PHOTOELECTRONS

Calculate the kinetic energy (in eV) of the photoelectrons emitted from lead by ultraviolet light with a frequency of 1.20×10^{15} Hz. The work function of lead is 4.14 eV.

Thinking	Working
Recall Einstein's photoelectric equation.	$E_{k\ max} = hf - \phi$
Substitute values into this equation.	$hf = \dfrac{6.63 \times 10^{-34}}{1.6 \times 10^{-19}} \times 1.20 \times 10^{15} = 4.97\ \text{eV}$ $E_{k\ max} = 4.97 - 4.14$ $= 0.83\ \text{eV}$

THE STRUCTURE OF ATOMS

(See Ch. 7.5 of Pearson Physics 12 WA.)

The Bohr Model

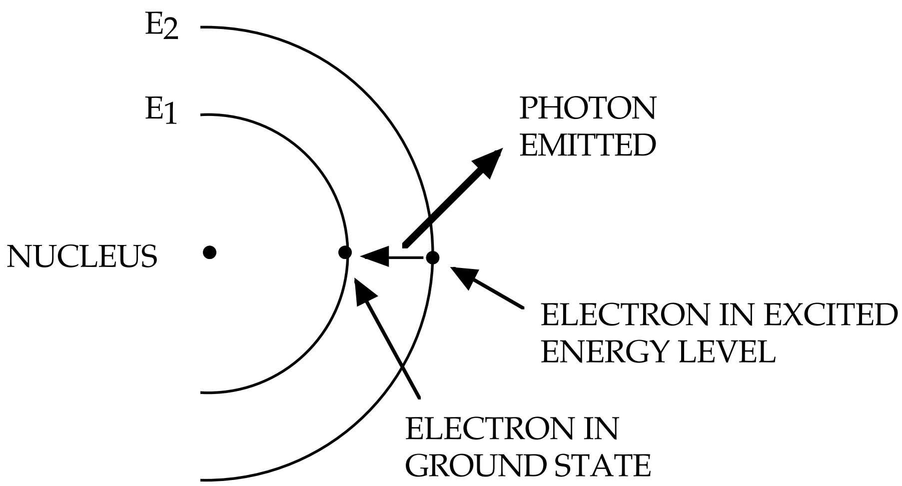
This model was used to improve an earlier attempt at describing an atom by Rutherford.

The Bohr model is based on the following ideas.

1. Electrons, which orbit the nucleus, can only have certain energies and move around in "energy levels". These levels are given numbers, the lowest level given number 1.

2. Electrons can only have the energy of the energy level that it was in.
   i.e. Only particular energy levels are possible.

3. Photons of energy are emitted as an electron moves from a higher energy level to a lower energy level.



At normal temperatures, electrons will fill the lowest energy levels possible. This is the ***ground state*** of an atom.

Bohr devised an equation that gives each of the energy levels in the hydrogen atom.

i.e.
$$\boxed{\begin{aligned} E_n &= \frac{-13.6\ \text{eV}}{n^2} \\ &= \frac{-2.17 \times 10^{-18}\ \text{J}}{n^2} \end{aligned}}$$

where n = the number of the energy level.

The value 13.6 eV is the ***ionisation energy*** of a hydrogen atom.

i.e. The energy required to totally remove an electron from the atom.

The energy is negative because it is necessary to add energy to the electron to free it from the atom.

A free electron from an atom is arbitrarily given an energy value of zero.

PRODUCTION OF LINE SPECTRA

If energy is absorbed by the atom (either by heating, absorption of electromagnetic radiation or bombardment by electrons), one or more electrons can be moved into a higher energy level. The atom is now called an ***excited atom***.

The amount of energy absorbed by the electrons in the atom is ***equal exactly to the energy levels above ground state.***

When the excited electrons move back to the lower energy levels, they give off an equivalent amount of energy as photons of electromagnetic radiation (visible light, UV radiation, IR radiation, X -rays, gamma rays).

When returning from higher energy levels, the electrons can transition back in one step or in several steps, releasing photons of differing frequencies.

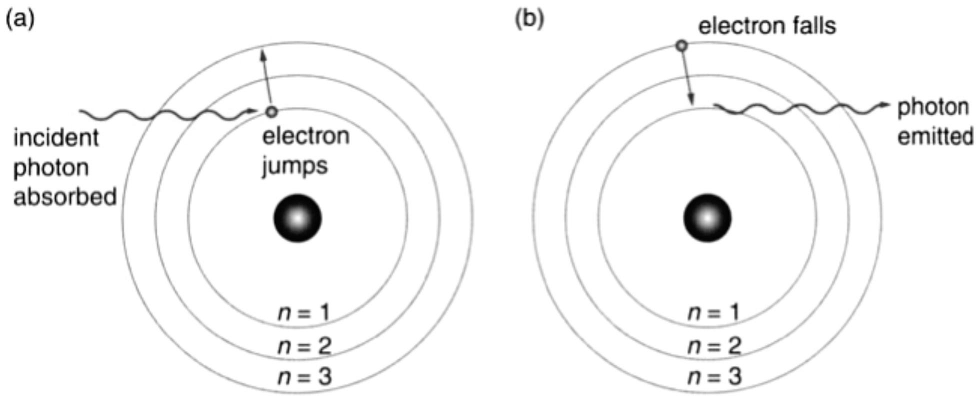
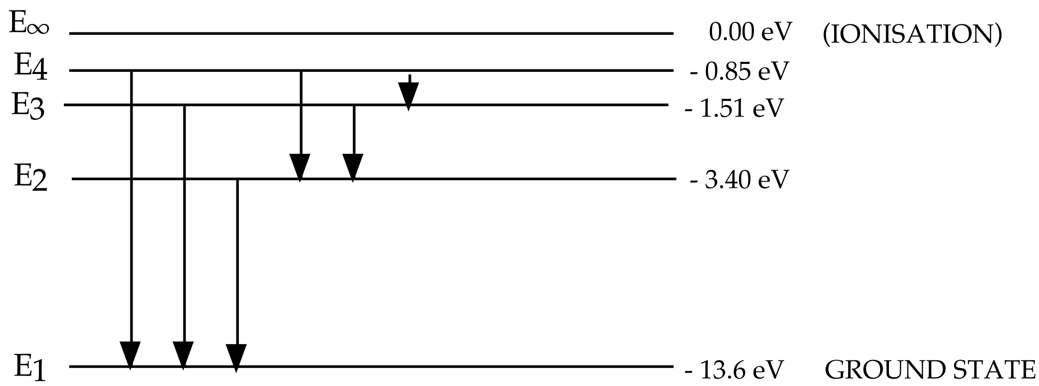


Figure 5.53

(a) If the incident photon carries a matching amount of energy, the electron can be knocked to one of the higher orbits as the photon is absorbed. The photon ceases to exist. (b) An atom will remain in an excited state for less than a millionth of a second. The electron will then fall to its ground state. The electron may fall in one step, or in a number of stages, emitting a photon(s) as it falls.

Example

Consider the energy levels of the hydrogen atom.



When an electron falls back to ground state from the fourth energy level, there are six possible transitions that can occur. Hence there are six observable spectral lines, each of a different frequency.

The particular wavelength and frequency of the radiation depends on the two particular energy levels involved.

The energy involved in the transition is given by:

$$\boxed{E_{photon} = |E_2 - E_1| = hf = \frac{hc}{\lambda}}$$

where E_{photon} = the energy released by the electron as it transitions back to its ground level (J)
E_2 = the upper energy level (J)
E_1 = the lower energy level (J)
h = Planck's constant = 6.62×10^{-34} Js
f = the frequency of the light emitted (Hz)
c = speed of light (ms^{-1})
λ = wavelength of the emitted photon of light (m)

Each of the emitted frequencies of light will correspond to a particular line in the spectral lines for the atom.

Example 1

An electron in an excited hydrogen atom drops from the third energy level (- 1.51 eV) to the second energy level (- 3.40 eV). Calculate the wavelength of the emitted photon.

Note 1.00 eV = 1.60 x 10^{-19} J

$$E_{photon} = |E_3 - E_2| = \frac{hc}{\lambda}$$

$$=> \quad [(-1.51) - (-3.40)]\,(1.60 \times 10^{19}) = \frac{(6.62 \times 10^{-34})(3.00 \times 10^{8})}{\lambda}$$

$$=> \quad \lambda = 6.567 \times 10^{-7}\ \text{m}$$

$$\therefore \quad \underline{\text{Wavelength} = 6.57 \times 10^{-7}\ \text{m.}}$$

Example 2

Five series of line spectra have been identified in the spectrum from an excited hydrogen atom. Three of these are the Lyman, Balmer and Paschen series.

From E = hf, the larger the energy transition, the larger the frequency of the emitted light. This corresponds to the UV end of the spectrum.

∴ Lyman series is in the UV range, Balmer series is in the visible range and the Paschen series is in the IR range.

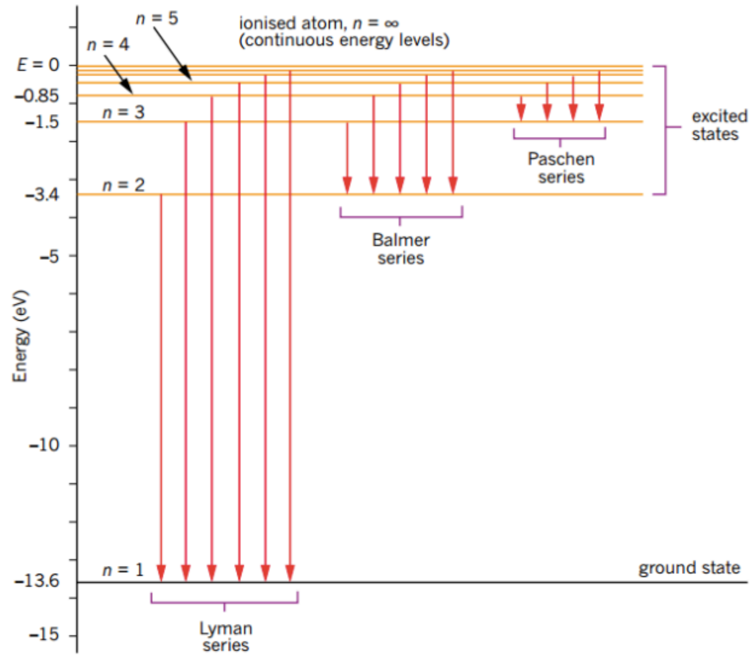
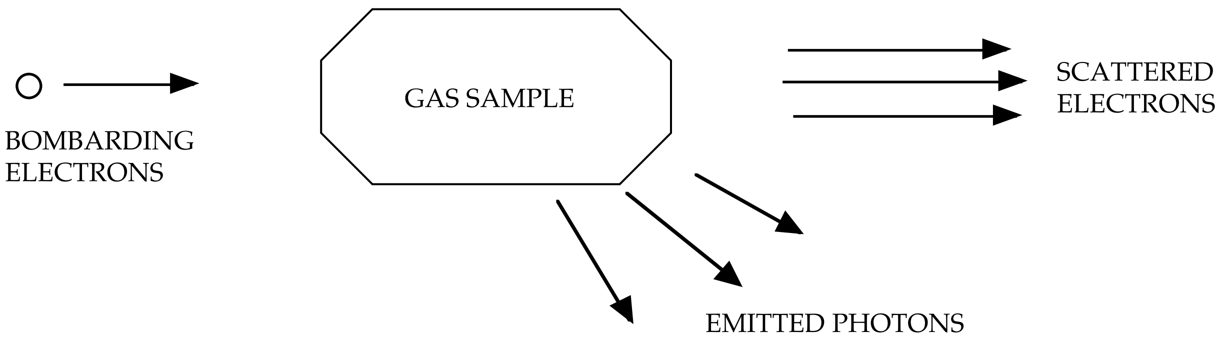


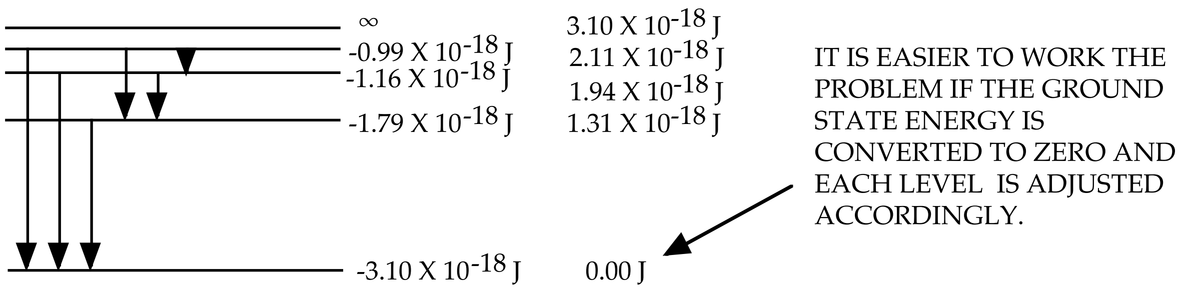
FIGURE 7.5.6 An energy level diagram for hydrogen. An electron in the ground state ($n = 1$) has an energy of (–)13.6 eV. For higher energy levels ($n > 1$), the energy levels can be seen to crowd together.

Example 3

A gas atom has the following energy levels - $E_1 = -3.10 \times 10^{-18}$ J, $E_2 = -1.79 \times 10^{-18}$ J, $E_3 = -1.16 \times 10^{-18}$ J, $E_4 = -0.99 \times 10^{-18}$ J. A sample of the gas is bombarded by a stream of electrons with energies of 2.25×10^{-18} J.

(a) Will the bombarding electrons be able to ionise the gas atoms? Explain.
(b) What will be the energy of the emitted photons?
(c) What will be the energy of the scattered electrons?





(a) The bombarding electrons have enough energy to reach the fourth energy level but not enough to reach the ionisation level.

(b) Having reached the fourth energy level, there are ***six*** possible energy transitions.

$E_4 - E_1 = 2.11 \times 10^{-18}$ J

$E_3 - E_1 = 1.94 \times 10^{-18}$ J

$E_2 - E_1 = 1.31 \times 10^{-18}$ J

$E_4 - E_2 = (2.11 - 1.31) \times 10^{-18}$
$= 0.80 \times 10^{-18}$ J

$E_3 - E_2 = (1.94 - 1.31) \times 10^{-18}$
$= 0.63 \times 10^{-18}$ J

$E_4 - E_3 = (2.11 - 1.94) \times 10^{-18}$
$= 0.17 \times 10^{-18}$ J

(c) The bombarding electrons will only lose an amount of energy ***equal to one of the energy levels*** of the gas atom, not an amount equal to the difference between energy levels.

i.e. The absorbed energies will be 1.31×10^{-18} J, 1.94×10^{-18} J, 2.11×10^{-18} J.

$\therefore$ Scattered energies will be as follows.

2.25×10^{-18} J (electrons go through the gas without hitting an atom)

$(2.25 - 1.31) \times 10^{-18} = 0.94 \times 10^{-18}$ J

$(2.25 - 1.94) \times 10^{-18} = 0.31 \times 10^{-18}$ J

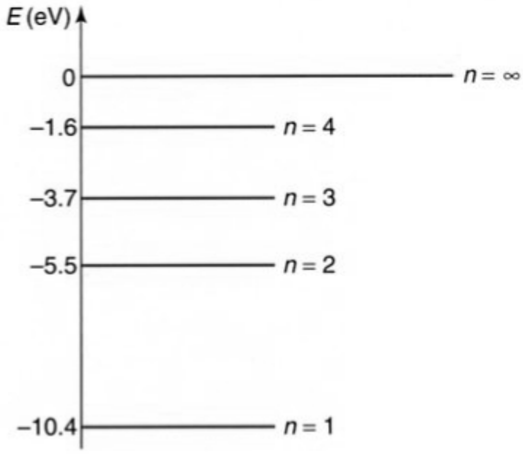
$(2.25 - 2.11) \times 10^{-18} = 0.14 \times 10^{-18}$ J

Example 4

The energy levels for atomic mercury are depicted in the diagram.

a Consider the mercury atom with its valence electron in the ground state. Ultraviolet light with photon energies 4.9, 5.0 and 10.50 eV is incident on some mercury gas. What could happen?

b Determine the wavelength of the light emitted after an electron in an excited mercury atom makes the transition from $n = 3$ to $n = 1$.



Solution

a The 4.9 eV photon may be absorbed, promoting the electron from the ground state to the first excited state. The 5.0 eV photon cannot be absorbed since there is no energy level 5.0 eV above the ground state. The 10.5 eV photon may ionise the mercury atom. In this case, the ejected electron will leave the atom with 0.1 eV of kinetic energy.

b First determine the difference in the energy levels in joules:

$E_3 = -3.7$ eV

$E_1 = -10.4$ eV

$$\Delta E = (E_3 - E_1) \times (1.60 \times 10^{-19})$$
$$= [(-3.7) - (-10.4)] \times (1.60 \times 10^{-19})$$
$$= 1.07 \times 10^{-18} \text{ J}$$

Then calculate the wavelength of the emitted photons:

$h = 6.67 \times 10^{-34}$ J s

$c = 3.00 \times 10^{8}$ m s^{-1}

$\Delta E = 1.07 \times 10^{-18}$ J

$$E_{photon} = hf = \Delta E$$
$$h\frac{c}{\lambda} = \Delta E$$
$$\lambda = \frac{hc}{\Delta E} = \frac{(6.67 \times 10^{-34})(3.00 \times 10^{8})}{1.07 \times 10^{-18}}$$
$$= 1.87 \times 10^{-7} \text{ m}$$

$\lambda = 187$ nm (in the ultraviolet)

Example 5

A 100.0 W light globe produces yellow-green light of wavelength 500.0 nm. Determine the number of photons released from the globe every minute.

Solution

The globe will release a total energy of:

$P = 100.0$ W

$\Delta t = 60.0$ s

$E = P\Delta t = (100.0)(60.0)$

$= 6.00 \times 10^{3}$ J

The frequency of the photon is:

$c = 3.00 \times 10^{8}$ m s^{-1}

$\lambda = 500.0 \times 10^{-9}$ m

$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8}}{500.0 \times 10^{-9}}$

$= 6.00 \times 10^{14}$ Hz

This energy is carried by N photons, and the energy of each photon will be given by:

$E = 6.00 \times 10^{3}$ J

$f = 6.00 \times 10^{14}$ Hz

$h = 6.63 \times 10^{-34}$ J s

$E = Nhf$

$N = \frac{E}{hf} = \frac{6.00 \times 10^{3}}{(6.63 \times 10^{-34})(6.00 \times 10^{14})}$

$= 1.51 \times 10^{22}$ photons

1.51×10^{22} photons emitted each minute—a very large number indeed!

Light-emitting Diodes (LEDs)

A ***light-emitting diode*** (LED) is a semi-conducting device that uses the excitation of electrons to produce light. LEDs are made from a variety of semiconductors, including GaAs, CdS, silicon and GaN-based materials. The addition of a small amount of another material, in a process known as 'doping', can change the conductive properties of semiconductors.

In a semiconductor diode, most of the electrons sit in an energy band known as the ***valence band***. At a slightly higher energy is another energy band, known as the ***conduction band***.

If a small amount of electrical energy is provided as a potential difference, electrons can jump from the valence band to the conduction band and then move through the semi-conductor as an electric current. When the electrons eventually drop back into the valence band, their energy is released as photons.

The difference in energy between the conduction band and the valence band level is known as the ***band gap***, E_g.

To turn on a basic LED, a potential difference or voltage needs to be applied. The threshold voltage, V_{th}, or potential difference required, is approximately equivalent to the band gap, E_g, or difference in the two energy levels of the emitting photons.

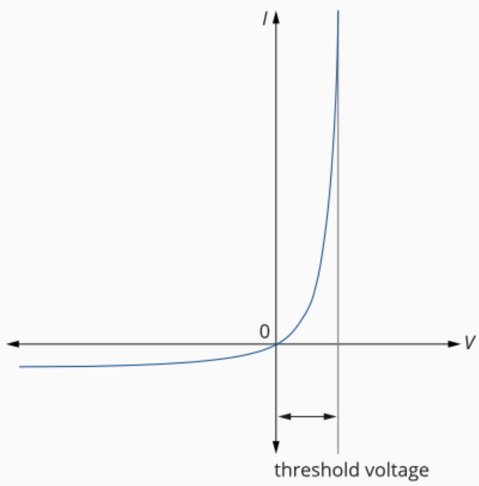
The colour or wavelength (λ) of the light produced is determined by the energy gap between the valence and conduction bands.

An approximate relationship can be given by:

$$\boxed{V_{th}q = \frac{hc}{\lambda} = E_g}$$

Measuring the threshold voltage for a series of LEDs can give an approximate measure of the band gap and emitting wavelength.

If the wavelength or frequency of the LED is known, then V_{th} can be plotted versus f. Comparing this to an equation of a straight line, $y = mx + c$, it can be seen that the slope of the line is hq. Knowing q, the electron charge, a value for h can be determined.



7.5.9 • I–V curve for an LED.

PHYSICS IN ACTION

Modern lighting

LEDs are gradually replacing most common uses of lighting, for example in cars, traffic lights and advertising signs. One of the newest forms of lighting on the market is white light LEDs. These use significantly lower currents than both tungsten filament lamps and incandescent lamps and tubes. In addition, they are very robust and typically have a lifetime of 20000 hours. Since the cost of production has significantly reduced, these lights are rapidly replacing conventional sources.

At the time of writing, these white light LEDs are made of a semiconducting alloy, indium gallium nitride, which emits light in the blue or ultraviolet range and is coated with a phosphor. The emitted light is absorbed by the phosphor, which then emits white light, similar to the way a fluorescent tube works; a simplified diagram is shown in Figure 7.5.13.

In the future, it is expected that white light LEDs will be made of a composite of three different semiconducting structures that will emit in the red, blue and green and thus mix to produce white light. (Take a ×10 magnifying lens and look at a white part of your computer screen to see light mixing in action.)

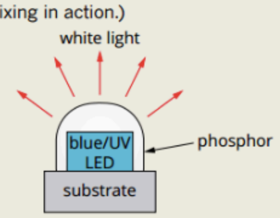


FIGURE 7.5.13 White light LEDs are currently made from an indium gallium nitride semiconductor alloy. The blue or ultraviolet light emitted interacts with a phosphor coating that radiates white light.

Limitations to Bohr's Model

- Bohr's Model can ***only be used to explain the spectral lines observed in the hydrogen atom***.
- ***It fails to explain the emitted spectral lines of any other gas***.

Quantum mechanics, a complex mathematical model developed by scientists (de Broglie, Schrodinger, Heisenberg), accounted for the results of experimental studies. It is based on the theory that particles have some wave characteristics, just as waves exhibit certain particle characteristics.

(See Ch. 7.6 of Pearson Physics 12 WA.)

In 1924, Louis de Broglie proposed a ground-breaking theory. He suggested that since light, which had long been considered to be a wave, sometimes demonstrated particle-like properties, then perhaps matter, which was considered to be made up of particles, might sometimes demonstrate wave-like properties.

He quantified this theory by predicting that the wavelength of a particle would be given by the equation:

$$\boxed{\lambda = \frac{h}{p}}$$

where p = the momentum of the particle ($kgms^{-1}$)
h = Planck's constant (= 6.63 x 10^{-34} Js)

This is also commonly written as:

$$\boxed{\lambda = \frac{h}{mv}}$$

where m = the mass of the particle (kg)
v = the velocity of the particle (ms^{-1})

PHYSICS IN ACTION

Solar sailing

In interplanetary space, where other forces such as friction are negligible, light pressure can actually be used as a form of propulsion. Spacecraft such as the Mariner 10 and MESSENGER spacecraft, which both flew past Mercury and Venus, used deceleration caused by solar pressure to conserve fuel.

More recently, the Japanese Aerospace Exploration Agency launched IKAROS (Interplanetary Kite-craft Accelerated by Radiation of the Sun). IKAROS (Figure 7.6.10) is the first spacecraft to draw its primary propulsion from a *solar sail*. A traditional sail propels a ship using the change of momentum that occurs when air molecules bounce off it. Similarly, a solar sail gains propulsion from changes in photon momentum as light is reflected from it. The IKAROS spacecraft has a 196 m^2 reflective sail that produces a thrust of 1.12 mN.

FIGURE 7.6.10 The IKAROS spacecraft is the first interplanetary spacecraft to use solar-sail technology.

De Broglie's prediction was confirmed by the Americans Davisson and Germer in 1927 when they observed diffraction patterns being produced in an experiment where they bombarded the surface of a piece of nickel with electrons.

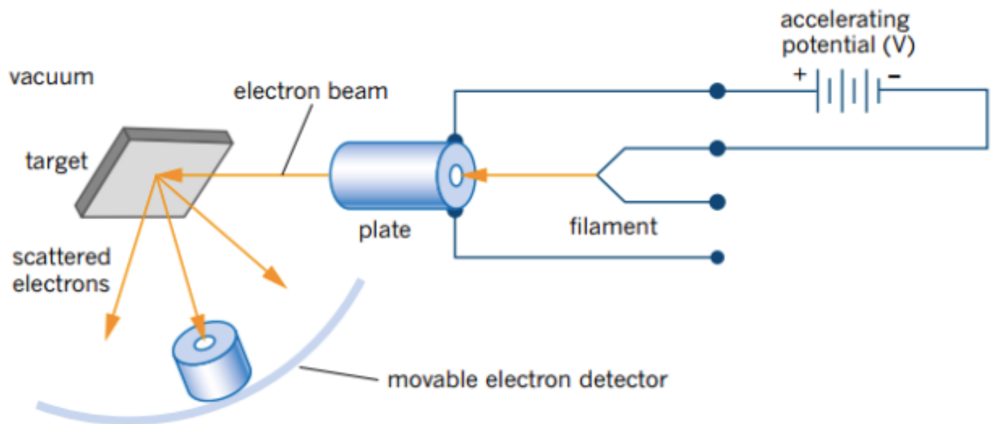


FIGURE 7.6.5 The Davisson and Germer apparatus to show electron scattering.

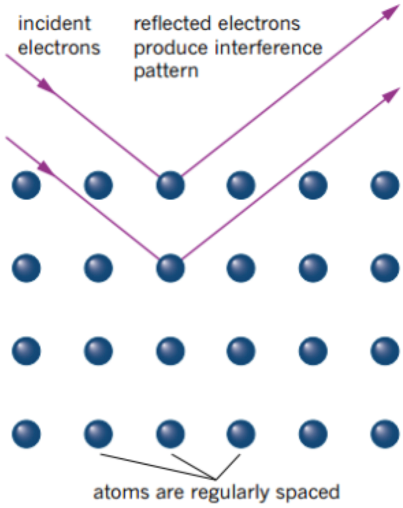


FIGURE 7.6.7 Electrons reflecting from different layers within the crystal structure create an interference pattern like those produced by a diffraction grating. If the path difference between the two rays is equal to a multiple number of wavelengths, constructive interference occurs; destructive interference occurs if the path difference is a multiple number plus half a wavelength.

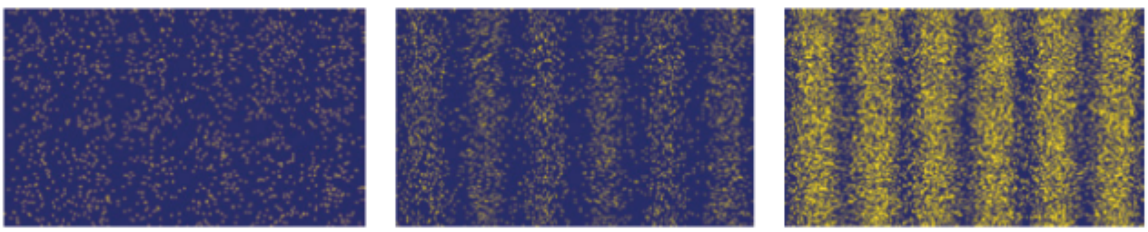


FIGURE 7.6.6 An electron diffraction pattern like the one observed by Davisson and Germer can be built up over time from repeated observations.

Wavelengths of Photons and Electrons

Also, in 1927, G.P. Thompson passed a beam of electrons and x-rays through the same crystal lattice. The wavelength of the x-rays was matched to that of the electrons. What he found was that the two diffraction patterns were almost identical, providing further proof that particles have wave properties.

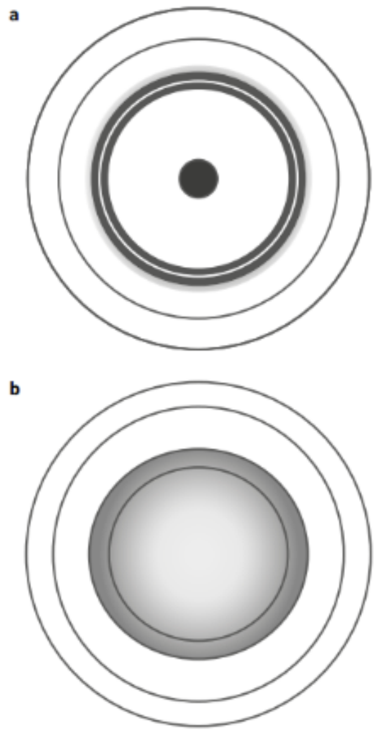


FIGURE 7.6.8 These diffraction patterns were taken by using (a) X-rays and (b) a beam of electrons with the same target crystal. Their similarity suggests a wave-like behaviour for the electrons and an electron de Broglie wavelength similar to that of X-rays.

EXTENSION

Electron microscopes

The discovery of the wave properties of electrons had an important practical application in the invention of the electron microscope. Just as an optical microscope makes use of the wave properties of photons to magnify tiny objects, so too can the wave properties of electrons be used to create magnified images (Figure 7.6.9).

One of the limitations of an optical microscope is that it can only create a clear image of structures that are similar in size to the wavelength of the light being used. This is because the light diffracts around these structures. So a light microscope is only useful for seeing things down to about 390 nm, the lower wavelength end of the visible light spectrum.

However, the wavelength of a beam of electrons is significantly smaller than the wavelength of a beam of visible light. For example, an electron microscope with electrons at an energy of 200 keV has a de Broglie wavelength of 2.51 pm (1 pm = 10^{-12} m). Due to lensing restrictions, the practical resolution limit for an electron microscope is 0.1 nm (1 nm = 10^{-9} m). Therefore, electron microscopes can create images with much finer detail than optical microscopes.

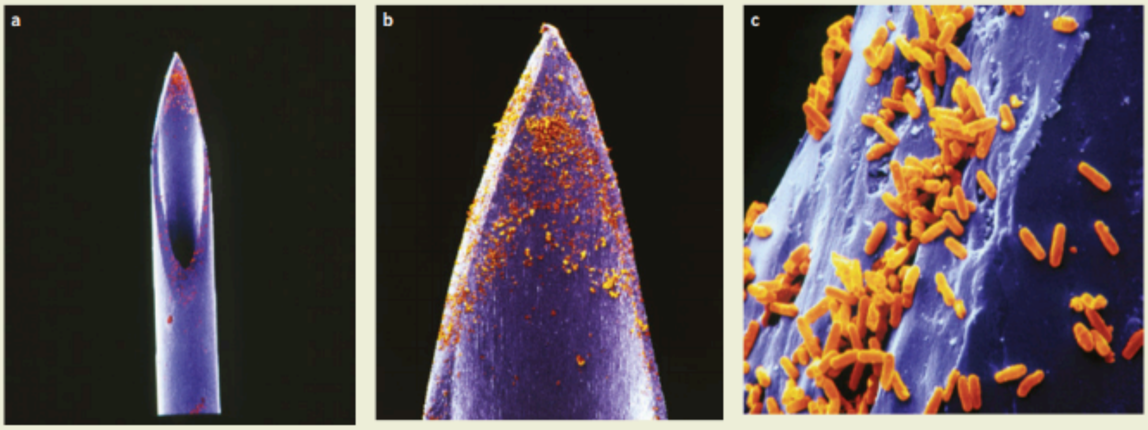


FIGURE 7.6.9 Images formed by an electron microscope: rod-shaped bacteria (orange) clustered on the point of a syringe used to administer injections. The magnifications are (a) ×9, (b) ×36 and (c) ×560 at 35 mm size.

Examples

1.

CALCULATING THE DE BROGLIE WAVELENGTH

Electrons in a famous experiment known as the Davisson–Germer experiment travelled at about $4.0 \times 10^6\,\mathrm{m\,s^{-1}}$. Calculate the de Broglie wavelength of these electrons given that the mass of an electron is 9.11×10^{-31} kg.	
Thinking	**Working**
Recall de Broglie's equation.	$\lambda = \dfrac{h}{mv}$
Substitute the appropriate values into the equation and solve it.	$\lambda = \dfrac{h}{mv}$ $= \dfrac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 4 \times 10^{6}}$ $= 1.8 \times 10^{-10}\,\mathrm{m}$ or $0.18\,\mathrm{nm}$

2.

CALCULATING THE DE BROGLIE WAVELENGTH OF A MACROSCOPIC OBJECT

Calculate the wavelength of a cricket ball of mass $m = 160\,\mathrm{g}$ travelling at $150\,\mathrm{km\,h^{-1}}$.	
Thinking	**Working**
Convert mass and velocity to SI units.	$m = 160\,\mathrm{g} = 0.16\,\mathrm{kg}$ $v = 150 \div 3.6 = 42\,\mathrm{m\,s^{-1}}$
Recall de Broglie's equation.	$\lambda = \dfrac{h}{mv}$
Substitute the appropriate values into the equation and solve it.	$\lambda = \dfrac{h}{mv}$ $= \dfrac{6.63 \times 10^{-34}}{0.16 \times 42}$ $= 9.9 \times 10^{-35}\,\mathrm{m}$

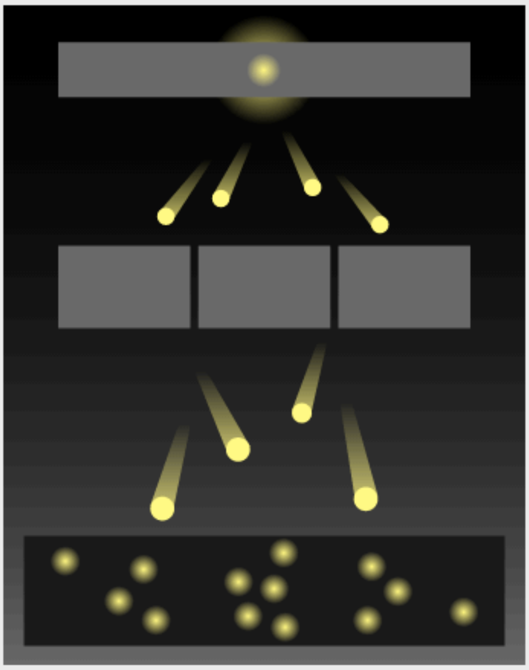
The quantum nature of matter

The quantum model of light (energy) was extended to matter. It led to a fundamental shift in the way we view the universe around us.

In the early years of quantum theory, some scientists believed that the wave properties of light observed in Young's double slit experiment may have been due to some sort of interaction between photons as they passed through the slits together.

Experiments were done with light sources that were so dim that scientists were confident that ***only one photon*** was passing through the apparatus at a time. These experiments produced identical interference patterns to those done with bright sources.

Interestingly, when a detector is used to measure which slit the photon passes through, the wave pattern disappears and the photon acts like a particle.



7.6.3 • An interference pattern can be built up over time by a series of single photons passing through Young's experiment, demonstrating the wave–particle duality of light.

Matter Waves and The Bohr Atom

De Broglie's work allowed one of the great mysteries of the atom to be solved. In the previous Rutherford model of the atom, an electron orbiting the nucleus of an atom should radiate energy and spiral inwards. But this was not observed, and scientists could not explain why not.

In formulating his model of the hydrogen atom, Bohr assumed that if an electron of mass *m* were moving with velocity *v* in an orbit with radius *r*, this orbit would be stable if it matched the condition:

$$\boxed{mvr = n\frac{h}{2\pi}}$$

where n = an integer

This can be rearranged to:

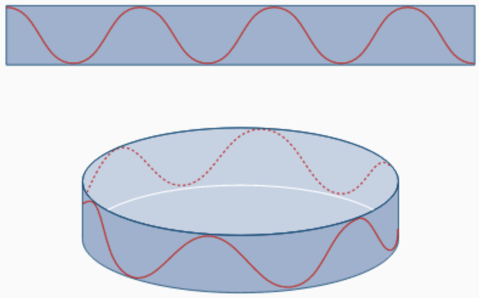
$$\boxed{2\pi r = n\frac{h}{mv}}$$

Since $2\pi r$ is the circumference C of a circle and by the de Broglie equation $\lambda = \frac{h}{mv}$, this equation can be rewritten as:

$$\boxed{C = n\lambda}$$

In other words, the ***stable orbits of the hydrogen atom are those where the circumference is exactly equal to a whole number of electron wavelengths***. An electron orbiting an atom is a type of electron ***standing wave***.

This can be visualised by imagining a conventional standing wave pattern, like in a vibrating string, being looped around on itself as shown in the figure below.



7.6.4 • A standing wave pattern (top) can be looped around on itself (bottom) if the circumference of the circle is equal to a whole number of wavelengths.

Erwin Schrödinger developed a mathematical equation that describes the wave behaviour of electrons. The wave properties of electrons are interpreted as representing the probability of finding an electron in a certain location.

Quantum mechanics is the name now given to the area of physics in which the wave properties of electrons are studied. Schrödinger's equation has been used to calculate the regions of space in which an electron can be found in a hydrogen atom. These are now known as 'orbitals' rather than orbits because they are ***complex three-dimensional shapes***, rather than the simple circular paths once imagined by Rutherford and Bohr.

Heisenberg's uncertainty principle

There are some quantities that are impossible to measure precisely. The first scientist to clearly identify this was the German physicist Werner Heisenberg. Heisenberg's uncertainty principle describes a limit to which some quantities can be measured.

For example, imagine we are trying to measure the exact location of an electron. The only way to do this would be to hit the electron with another particle such as a photon of light. However, as soon as the photon strikes the electron, it would cause the electron to move. In this way, the act of measurement causes a change in the value being measured. This is a general problem when trying to measure the location or motion of sub-atomic particles.

Scientists still argue about the interpretation of this equation, whether it means there is a fundamental 'fuzziness' to the universe, or if this just represents the limit of our ability to measure.

EXTENSION

Heisenberg's uncertainty principle

There are some quantities that are impossible to measure precisely. The first scientist to clearly identify this was the German physicist Werner Heisenberg (Figure 7.6.15). Heisenberg's uncertainty principle describes a limit to which some quantities can be measured.

For example, imagine you are trying to measure the exact location of an electron (Figure 7.6.16). The only way to do this would be to hit the electron with another particle such as a photon of light. However, as soon as the photon strikes the electron, it would cause the electron to move. In this way, the act of measurement causes a change in the value being measured. This is a general problem when trying to measure the location or motion of subatomic particles.

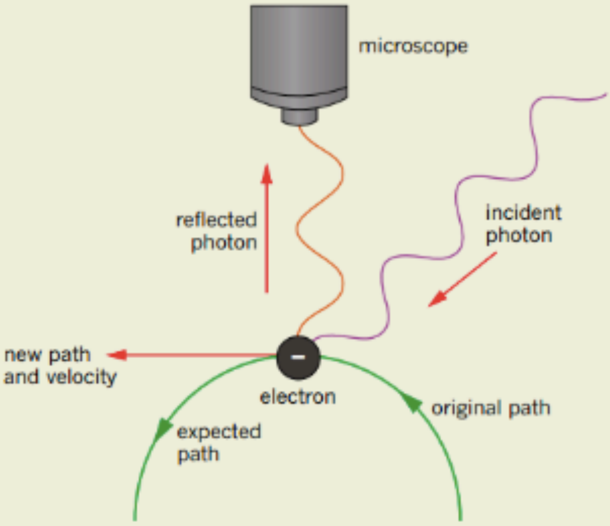


FIGURE 7.6.16 A thought experiment considering how an electron could be observed. The reflection of the photon needed to observe the electron introduces uncertainty in the position of the electron, making it unobservable.

Heisenberg described this problem with a formula that states that the product of the uncertainties in the position (Δx) and momentum (Δp) of a particle must always be greater than a certain minimum value related to Planck's constant (h):

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

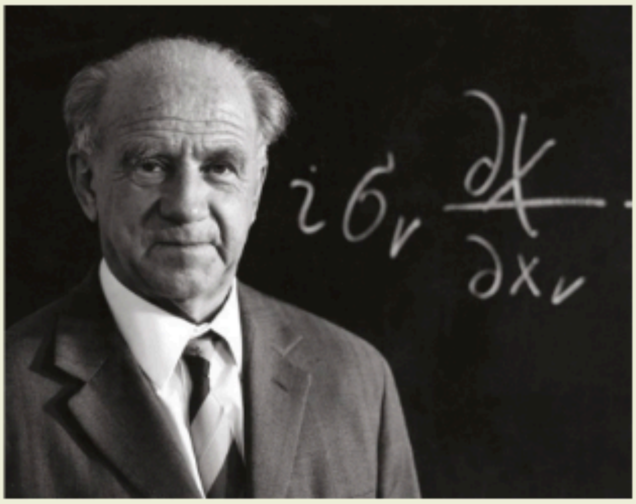


FIGURE 7.6.15 Werner Heisenberg (1901–76) won the Nobel Prize in Physics in 1932 for his work on the uncertainty principle. Heisenberg is regarded as a founder of quantum physics.

According to Heisenberg's uncertainty principle, the more exactly the position of a subatomic particle is known, the less is known about its momentum. Similarly, the more precisely the momentum of a particle is measured, the less certain is its exact position.

It can also be expressed as $\Delta E \Delta t \geq \frac{h}{4\pi}$, where ΔE is the uncertainty in energy and Δt the uncertainty in time

For the normal-sized world around us, the inclusion of Planck's constant, h, in the measure of uncertainty means that the level of uncertainty in determining the position of everyday objects is extremely small—in fact, virtually insignificant.

However, at an atomic scale, this level of uncertainty is substantial. And since everyday objects are made up of atoms containing subatomic particles such as electrons, the basic understanding of matter comes down to this fundamental property of all quantum mechanical systems.

Single-slit diffraction and the uncertainty principle

An experiment that can be used to illustrate Heisenberg's uncertainty principle is the single-slit diffraction of light or electrons.

If light from a laser is shone through a narrow adjustable slit, a diffraction pattern will form on the screen behind the slit (Figure 7.6.17), and if single photons pass through the slit, a diffraction pattern is also formed.

The observations can be explained by treating the laser light as a wave and considering the interference of different sections of the wavefront that pass through the slit. However, explaining this pattern in terms of the motion of individual photons is much more challenging.

Consider a single photon on its journey from the laser to the screen. As the photon passes through the slit, its position is known to some degree of certainty (Δx). According to Heisenberg, this means that there must therefore be an uncertainty about the photon's momentum (Δp), which is why the photon can end up at a variety of different places on the screen. Schrödinger's wave equation could be used to explain why some paths for the photon are much more likely than others.

Since the slit is adjustable, it could be made narrower. This would decrease the uncertainty about the photon's position, Δx, but increase the uncertainty about the photon's momentum, Δp. This is why narrowing the slit causes the diffraction pattern to spread out.

If the slit is removed, as shown in Figure 7.6.18, there are no constraints on the path of the light, the uncertainty about the position (Δx) of each individual photon is large. Correspondingly, the uncertainty about the momentum of the photon becomes very small and no diffraction occurs—all the photons end up very close together in a small spot in the middle of the screen.

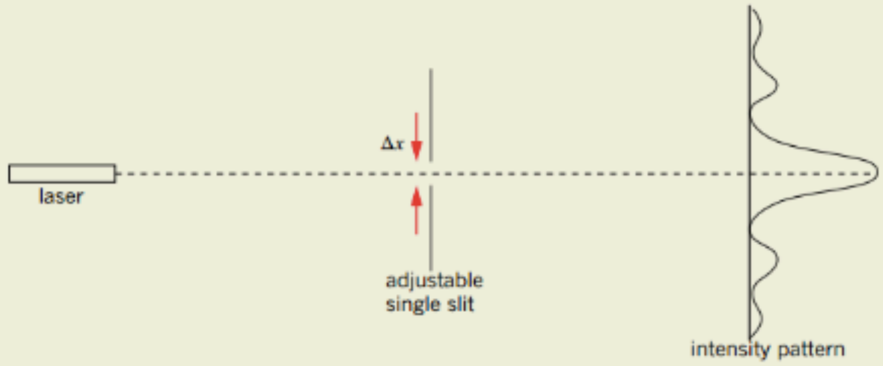


FIGURE 7.6.17 Demonstrating Heisenberg's uncertainty principle with diffraction through a single slit.

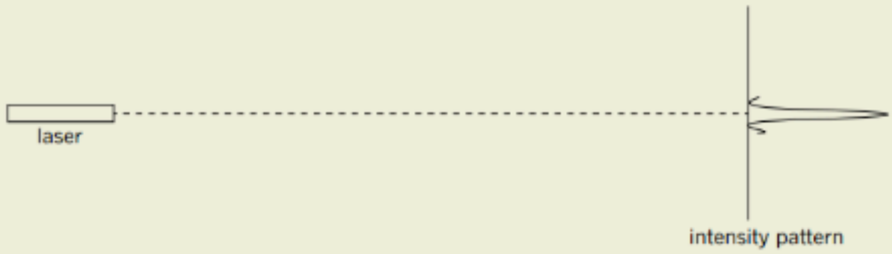


FIGURE 7.6.18 Without a slit, there is no uncertainty about the momentum of the photon and therefore no diffraction occurs.

CLASSIFICATION OF SPECTRA

(See Ch. 7.5 of Pearson Physics 12 WA.)

There are two classes of spectra - ***emission and absorption spectra***.

Emission spectra are obtained by the dispersion of light coming directly from a source due to its bombardment by electrons or by heating.

Absorption spectra occur when light is passed through an absorbing material.

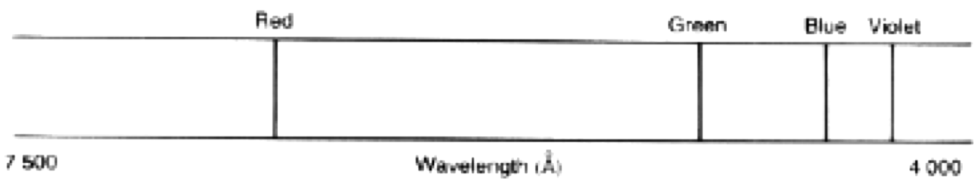
Emission Spectra

1. ***Line Emission Spectra***

These are produced by emissions from ***gases under low or normal pressures***.

The spectrum is a number of single, sharply-defined lines on a dark background.

Each element has its own characteristic line spectrum. It is due to this that line emission spectra can be used to identify the constituents of unknown substances.



The hydrogen spectrum — a line emission spectra

2. ***Band Emission***

These are produced by ***excited molecules in the gaseous state***. Molecules have many more energy levels than atoms and the energy levels are in close groups.

The spectrum is made up of bands with a sharp edge and a diffuse edge on a dark background.

It is actually made up of a group of lines that come closer together as they approach the sharp edge of the band.

Bands are present in the emission spectra produced from ***flames***.

Around the flame is a thin layer of gas (called a mantle) which contains combustion products such as CO and H_2O.

It is these molecules, excited by the heat of the flame, that account for bands in the emission spectra.

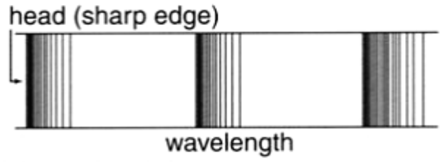


Figure 5.55

A typical band emission spectrum. Each band of colour is actually a large number of closely spaced lines. Typically, each band has a sharp edge where the lines are closely spaced and a fuzzy edge where the lines begin to spread out.

3. ***Continuous Emission Spectra***

These spectra consist of ***all colours from red at one end to violet at the other***, as all wavelengths of light are present.

This spectrum is produced by ***incandescent solids*** such as the tungsten filament in a light globe, or hot liquids such as molten iron in a smelter.

This type of radiation is called ***black-body radiation*** and the specific characteristics of the continuous spectrum depends on the temperature of the emitting solid or liquid.

At a temperature of around 1000 °K, a solid will glow 'red hot'. The filaments of light globes operate at around 3000 °K and glow 'white hot'. The light from incandescent light globes is rich in orange, red and infrared.

At around 6000 °K a black body emits a spectrum very much like the spectrum of the Sun. Sunlight is mainly red, orange and yellow.

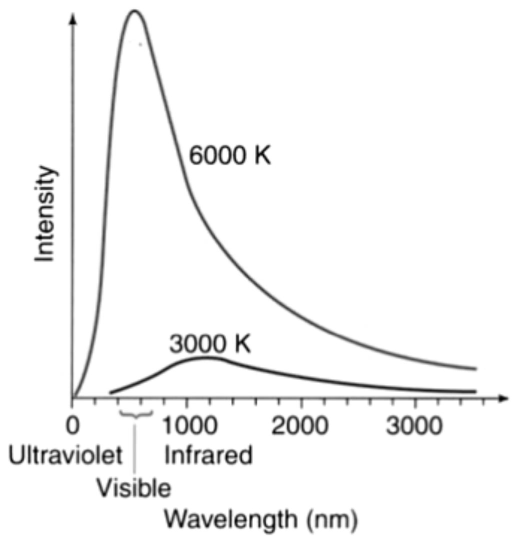


Figure 5.58

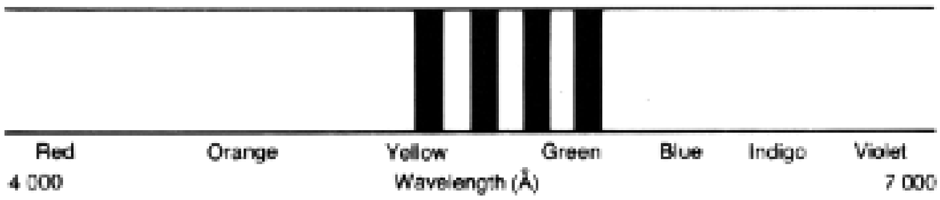
The continuous spectra emitted by a hot body at two different temperatures. At higher temperatures, the distribution includes a greater proportion of photons of higher energies and the total intensity is greater.

Absorption Spectra

1. ***Band Absorption***

These are produced when white light passes through a ***solid such as coloured glass**, or* ***liquid such as chlorophyll or potassium permanganate***.

Dark bands on a background of a continuous spectrum are seen.

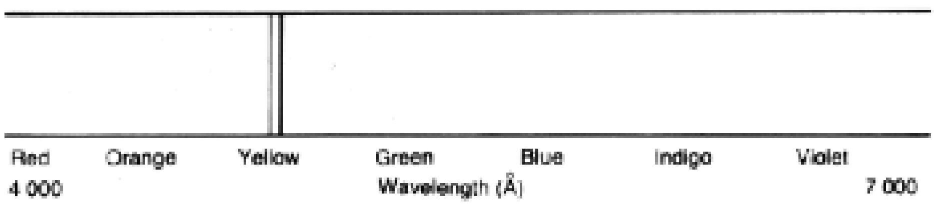


Band absorption spectrum for potassium permanganate solution

2. ***Line Absorption***

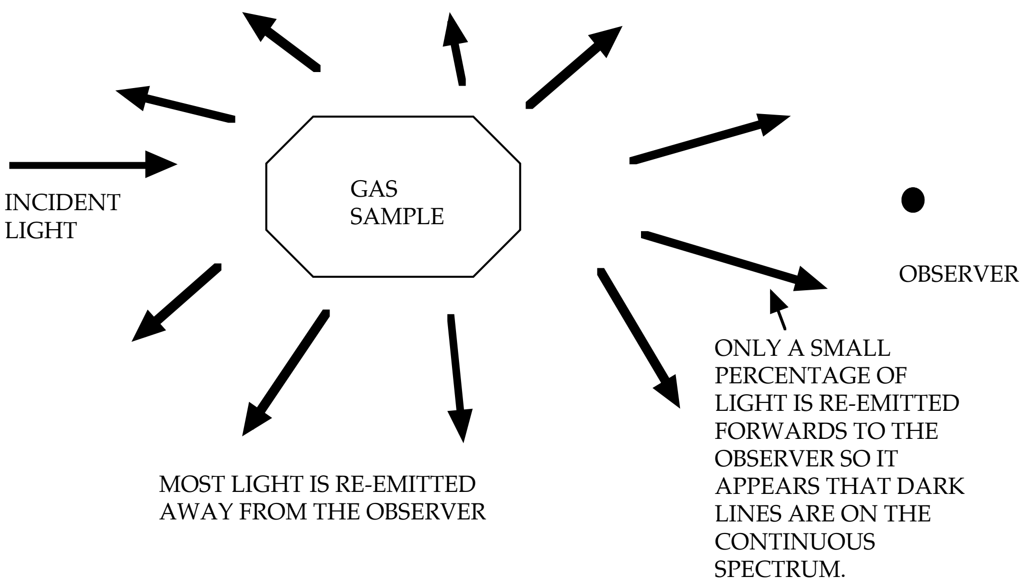
These are produced when white light passes through a ***vapour or gas***, producing dark lines on a continuous spectrum.

Black lines indicate which wavelengths have been absorbed



Line absorption spectrum for sodium vapour.

The lines ***are dark and not black*** because the light is absorbed and then re-emitted in all directions, so that only a small percentage is transmitted forwards to the observer or detector.



Remember that only certain frequencies of light are absorbed by the gas and the energy of these frequencies is ***exactly equal to the energy levels in the atoms in the gas above the ground state***.

3. Solar Spectrum

The solar spectrum consists of a continuous emission spectrum crossed by many absorption lines.

These lines are called ***Fraunhofer Lines***.

They are due to selective absorption of certain frequencies of light as white light from the interior of the Sun pass through the relatively cool vapours surrounding the Sun and the Earth's atmosphere.

Comparing the absorption spectrum with known emission spectra of elements has revealed that elements such as hydrogen, helium, sodium, iron and calcium are present in the Sun.

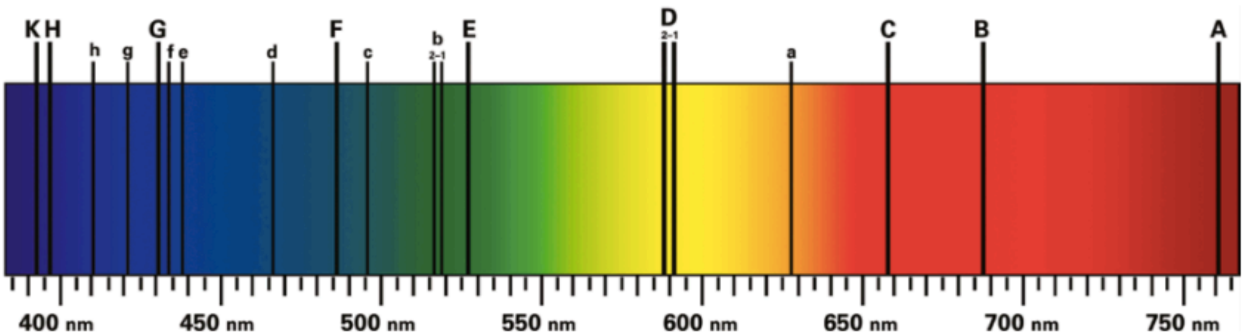


FIGURE 7.5.2 The spectrum of sunlight contains some missing colours known as Fraunhofer lines.

FLUORESCENCE AND PHOSPHORESCENCE

Fluorescence

A substance that is fluorescent appears to emit its own light ***but is not hot***.

The ability of an atom to fluoresce depends on the energy levels in the atom.

When the atom is excited from one energy state to a higher one by the absorption of a photon or by collision with an electron, it may return to its original state (ground state) by a series of two or more jumps (***provided there is at least one energy level in between***).

Generally, fluorescence occurs by the absorption of ultraviolet (UV) light.

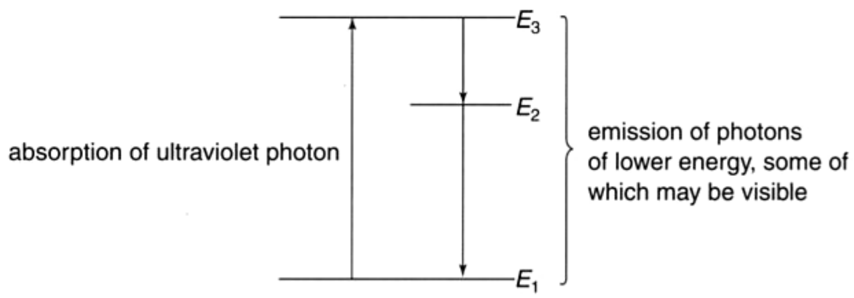


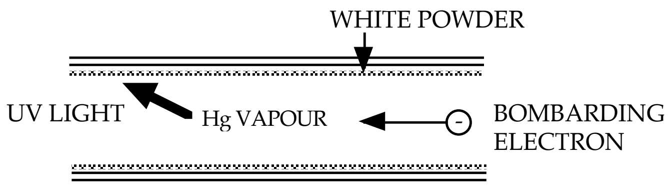
Figure 5.59
The process of fluorescence. An electron excited by the absorption of a high energy photon may then return to its ground state in more than one step, releasing photons of visible light.

The energy of the UV photon absorbed must be exactly equal to the difference between the ground state energy and the higher energy level.

By "falling back" in two or more jumps, the energy emitted is in smaller quantities and these correspond to frequencies in the visible region.

Examples

1. ***Fluorescent Tube***



A stream of electrons moves through a mercury vapour in the tube.

99% of the emitted energy from the Hg atoms is UV light, the rest is visible light.

This is absorbed by the white powder on the inside of the tube. It fluoresces and re-emits the energy as visible light.

Since glass will not transmit UV radiation, the glass bulb or envelope of a fluorescent tube will absorb the UV radiation that the fluorescent coating does not convert to visible light. The absorbed energy will appear as heat.

2. *Optical Brighteners*

A practical use of this phenomenon is the addition of fluorescent "whiteners" to laundry detergent.

These chemicals absorb ultraviolet light and re-emit the energy as blue light. Sunlight is mainly red, orange and yellow light, giving a "yellowish" light. White colours look "yellowish" as well as they reflect these colours.

By adding the blue light from the whiteners, the whites reflect all colours and look white. This has the visual effect of brightening the fabric.

3. Highlighters

The substance in these pens fluoresces and gives off a characteristic colour such as green, blue, orange, etc.

Phosphorescence

Some substances, when excited by photon absorption, form ***metastable states*** and don’t re-emit the energy for some time. The electrons can take seconds or even hours to drop back to their ground state so light is emitted for some time.

The process is identical to fluorescence but s called ***phosphorescence*** when there is a significant time delay involved.

Phosphorescent substances are said to be luminous and are used in applications such as the ***paint for watch dials*** and the ***coatings on television screens***.

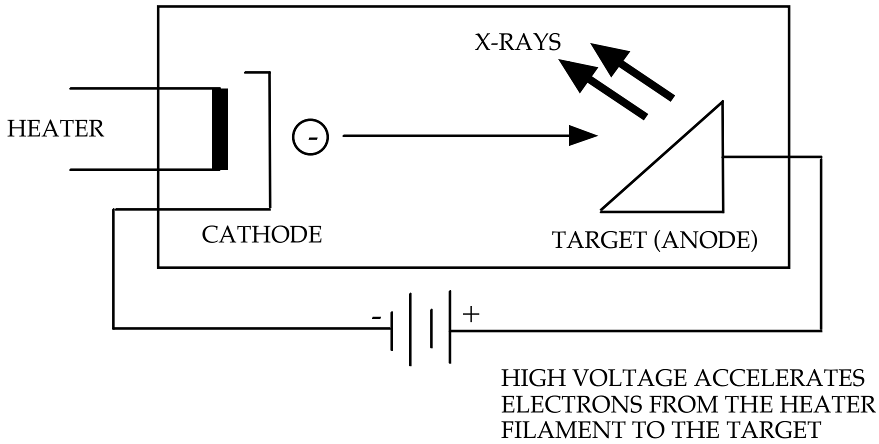
X - RAYS

An X-ray tube consists of a few basic components.

1. A heater produces electrons from the filament (called ***thermionic emission***).

2. An electric field (produced by high voltages of about 50.0 kV or more) accelerates electrons to very high velocities from the filament to the target.

3. Electrons decelerate rapids on hitting the target, giving some of their energy up as X-ray (~ 5% max) and the rest as heat.



At impact, the electrons have about 5.00×10^4 eV of E_k (calculated using $W = Vq = 1/2\ mv^2$).

The metal atoms rapidly decelerate the electrons and they lose their energy as X-ray photons and heat.

Example

Calculate the maximum frequency of an X-ray produced when an electron of 50.0 keV collides with a tungsten target and ***loses all of its energy*** as an X-ray.

$$E_k = (5.00 \times 10^4)(1.60 \times 10^{-19})$$
$$= 8.00 \times 10^{-15}\ \text{J}$$

$$E = hf$$
$$\Rightarrow \quad f = \frac{(8.00 \times 10^{-15})}{(6.62 \times 10^{-34})}$$
$$= 1.208 \times 10^{19}\ \text{Hz}$$

$$\therefore \quad \underline{\text{Maximum frequency} = 1.21 \times 10^{19}\ \text{Hz.}}$$

When the electrons strike the target, they can lose their energy in a number of ways.

- If an electron collides with the target and stops suddenly, it loses all of its kinetic energy at once and emits a single photon equal to the kinetic energy of the electron.

  Hence, the most energetic photon that can be obtained from an X-ray tube has a kinetic energy equal to the most energetic electrons accelerated by the accelerating voltage.

(See the above example.)

- When electrons strike the target, they will typically slow down in a number of collisions with individual atoms and lose their energy in more than one step.

  Consequently, they will create a number of photons each with less energy than the maximum possible photon energy.

  This explains why X-ray spectra are continuous and have a wide range of photon energies up to a maximum value.

Thus, a stream of electrons, all with the same energy, will produce a spectrum of X-ray photons with many different frequencies.

This spectrum is called **Bremsstrahlung radiation**.

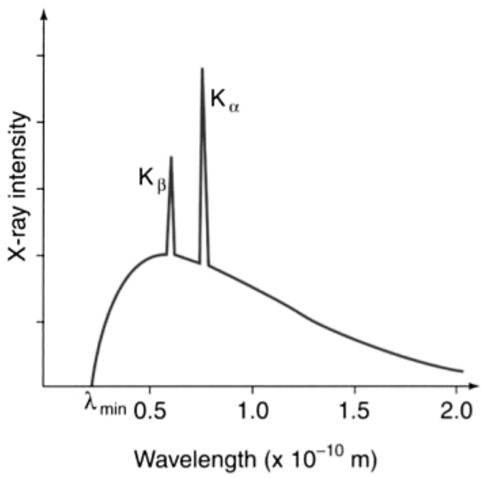


Figure 5.65
The spectrum obtained from a 50 kV X-ray tube with a molybdenum target anode.

The metal target atoms will also emit a few specific X-rays due to the incoming electrons knocking **inner electrons** out of the atoms.

An upper energy level electron drops down to take the place of the inner electron and consequently emits a high energy X-ray photon (due to the large energy transition).

Thus, a few peaks are added to the X-ray spectrum and are characteristic for that particular metal in the target.

The presence of these peaks can be used to identify metal samples as each metal produces a characteristic "finger print" with peaks corresponding to certain wavelengths.

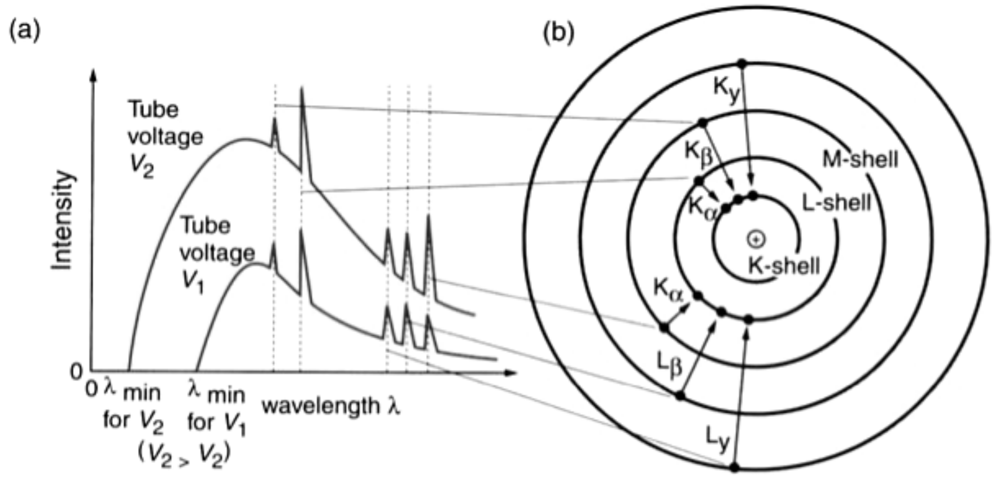


Figure 5.68

The spectrum produced by an X-ray tube depends on the accelerating voltage and the element in the target. λ_{min} depends on the maximum energy of the bombarding electrons. The peaks in the spectra are a line spectrum produced when inner orbital electrons are displaced and replaced by electrons from the higher energy shells.

Example

Figure 5.65 shows the spectrum of X-rays that would be obtained from a 50.0 kV X-ray tube with a molybdenum target.

a Calculate the maximum kinetic energy of the bombarding electrons in both electronvolts and joules.
b Calculate the maximum energy of the emitted photons in joules.
c Calculate the maximum speed of the bombarding electrons.
d Calculate the maximum frequency of the emitted X-rays.
e Calculate the minimum wavelength of the emitted photons.
f Calculate the photon energy of the K_α emission in joules and electronvolts.

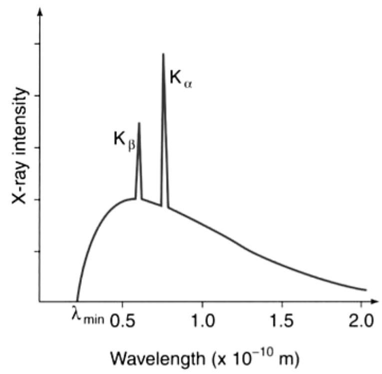


Figure 5.65
The spectrum obtained from a 50 kV X-ray tube with a molybdenum target anode.

Solution

a The accelerating voltage is 5.00×10^4 V so the maximum possible energy of the accelerated electrons is 5.00×10^4 eV. In joules this is:

$$E_{k(max)} = (5.00 \times 10^4) \times (1.60 \times 10^{-19})$$
$$= 8.00 \times 10^{-15} \text{ J}$$

since 1 eV = 1.6×10^{-19} J.

b The maximum energy of the emitted photons is the same as the maximum kinetic energy of the bombarding electrons:

$$E_{ph(max)} = E_{k(max)}$$
$$= 8.00 \times 10^{-15} \text{ J}$$

c The maximum speed of the bombarding electrons is found from the maximum kinetic energy:

$m_e = 9.11 \times 10^{-31}$ kg $\qquad E_{k(max)} = \frac{1}{2} m v^2_{max}$

$E_{k(max)} = 8.00 \times 10^{-15}$ J $\qquad v_{max} = \sqrt{\dfrac{2E_{k(max)}}{m}} = \sqrt{\dfrac{2(8.00 \times 10^{-15})}{9.11 \times 10^{-31}}}$

$$= 1.33 \times 10^8 \text{ m s}^{-1}$$

d The maximum frequency of the X-rays is given by:

$h = 6.63 \times 10^{-34}$ J s $\qquad E_{ph(max)} = hf_{max}$

$E_{ph(max)} = 8.00 \times 10^{-15}$ J $\qquad f_{max} = \dfrac{E_{ph(max)}}{h} = \dfrac{8.00 \times 10^{-15}}{6.63 \times 10^{-34}}$

$$= 1.21 \times 10^{19} \text{ Hz}$$

e The wavelength can be found from frequency and the speed of light using the wave equation:

$f_{max} = 1.21 \times 10^{19}$ Hz $\qquad \lambda_{min} = \dfrac{c}{f_{max}} = \dfrac{3.00 \times 10^8}{1.21 \times 10^{19}}$

$c = 3.00 \times 10^8$ m s^{-1} $\qquad = 2.49 \times 10^{-11}$ m

f The K_α line appears to correspond to a wavelength of 0.75×10^{-10} m

$\lambda_\alpha = 0.75 \times 10^{-10}$ m $\qquad E_\alpha = \dfrac{hc}{\lambda_\alpha} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{0.75 \times 10^{10}}$

$c = 3.00 \times 10^8$ m s^{-1} $\qquad = 2.65 \times 10^{-15}$ J

$h = 6.63 \times 10^{-34}$ J s $\qquad = 1.66 \times 10^4$ eV